Morphological, Anatomical and Physiological Adaptation of Birds

Vertebrate Circulatory Systems

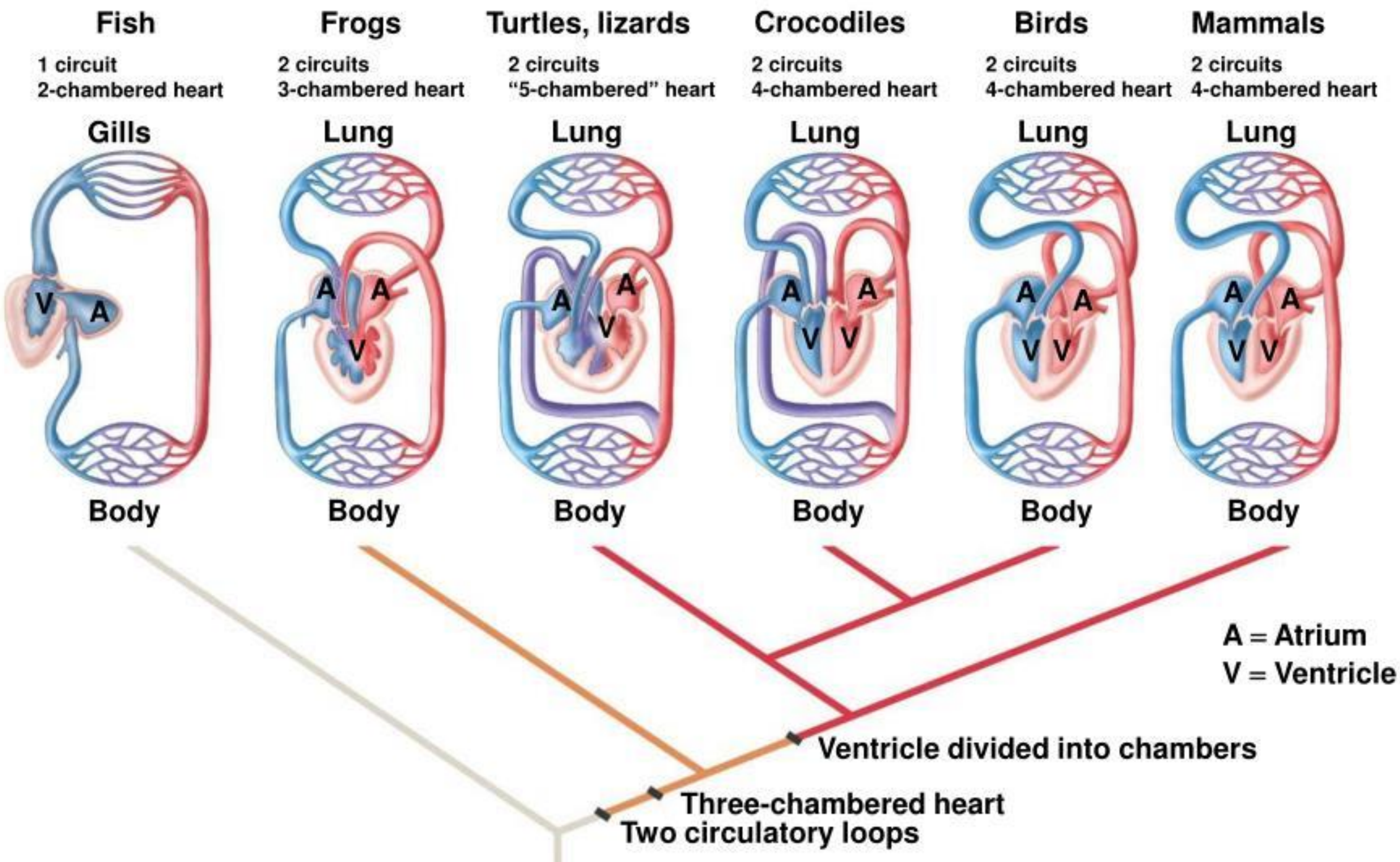
- The principal difference in vertebrate systems is the separation of the heart into two pumps.

- These changes occurred as vertebrates converted from gill breathing to lungs.

- In fish, oxygenated blood is provided to the body organs before the veins return to the heart.

- Terrestrial animals evolved lung breathing and eliminated gills between heart and aorta.

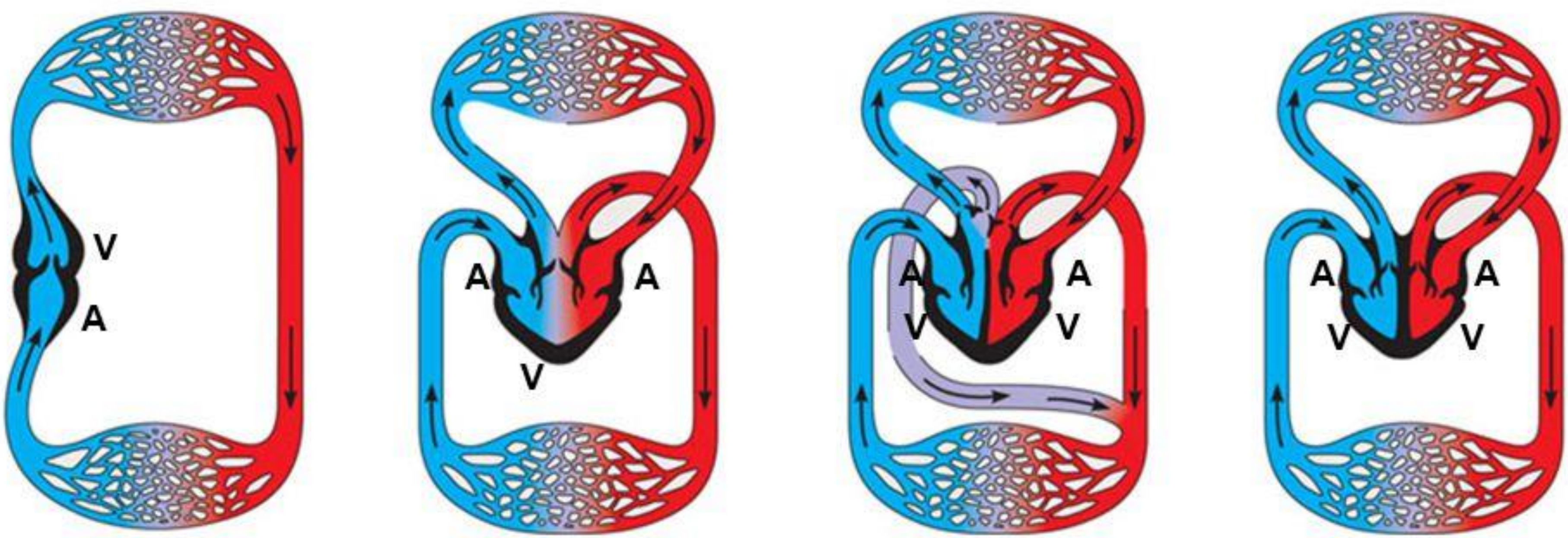
The Evolution of the Vertebrate Circulatory System

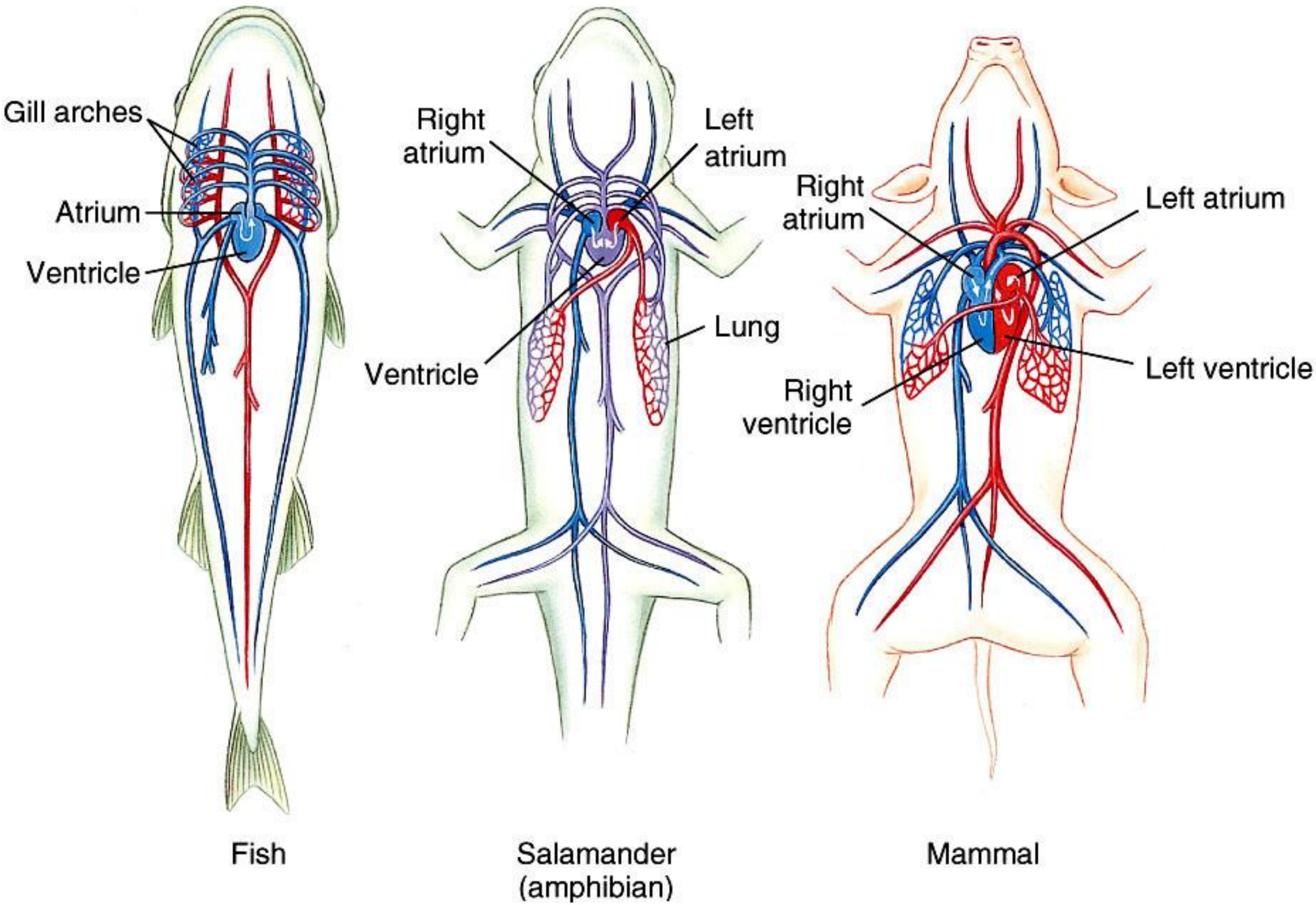


Evolution of circulatory system

Not everyone has a 4-chambered heart

fish	amphibian	reptiles	birds & mammals
2 chamber	3 chamber	3 chamber	4 chamber




Gill arches
Atrium
Ventricle
Fish
Right atrium
Left atrium
Ventricle
Lung
Salamander (amphibian)
Right atrium
Left atrium
Right ventricle
Left ventricle
Mammal

Double Circulation

- This provided a high pressure system that provided oxygenated blood to capillary beds and a pulmonary circuit to serve the lungs.

- This change is seen in lungfishes and amphibians.
- Modern amphibians have separate atria.
  - The right atrium receives venous blood from the body.
  - The left receives oxygenated blood from the lungs.
  - The ventricle is undivided but venous and arterial blood do not heavily mix.

- Ventricles are nearly separate in crocodilians and completely separate in birds and mammals.

Circulation in Birds

- The four-chambered heart is large, with strong ventricular walls.

- Birds share with mammals a complete separation of respiratory and systemic circulations.

- The right aortic arch, instead of the left as in mammals, leads to the dorsal aorta.

- The heartbeat is relatively fast compared to mammals and is inversely proportional to size.
  - A turkey heart beats 93 times per minute.
  - A chicken heart beats 250 times per minute.
  - A small black-capped chickadee heart beats 500 times per minute.

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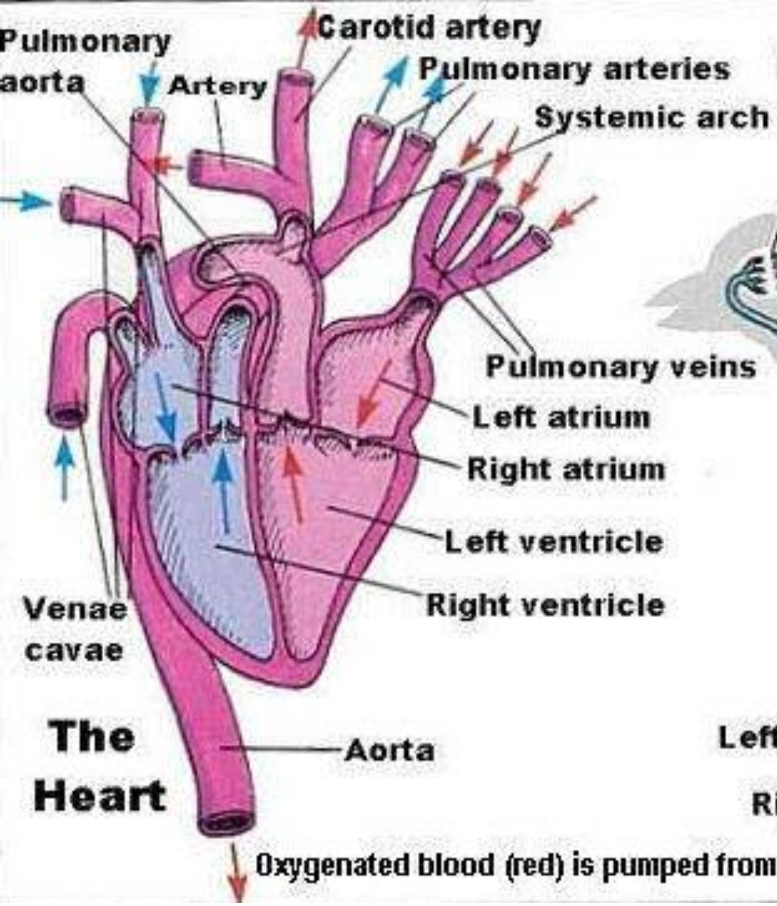
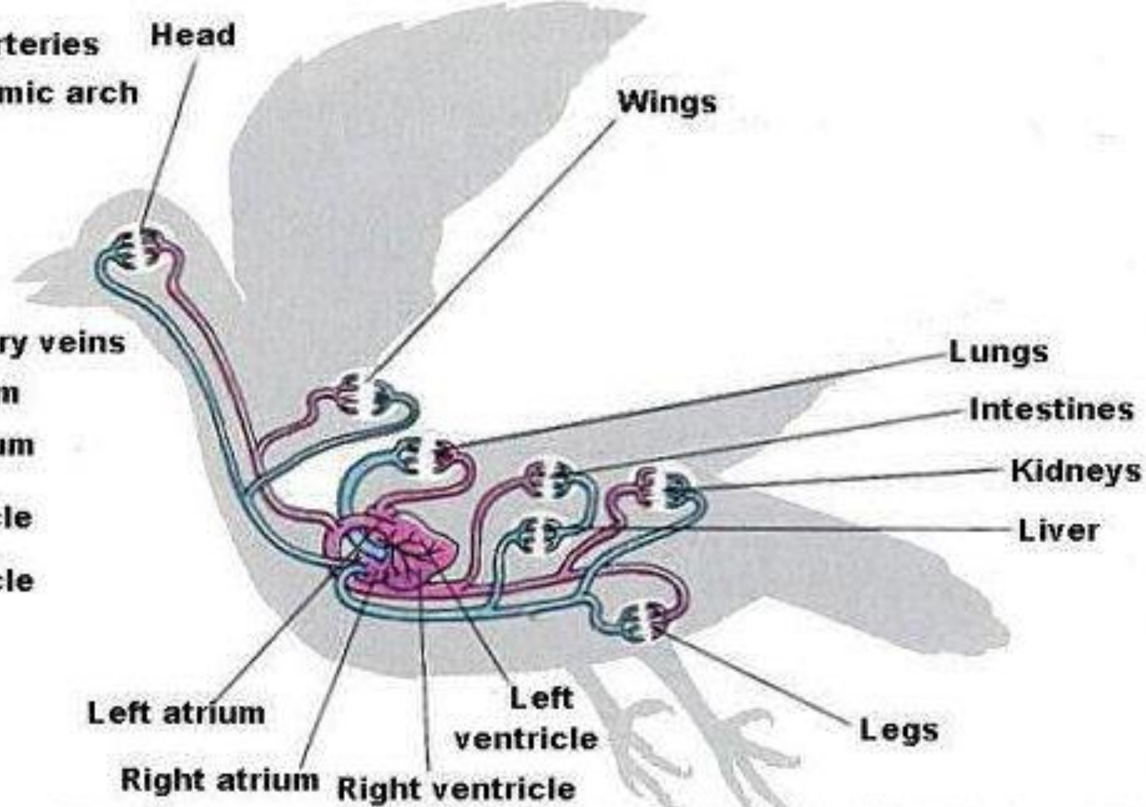
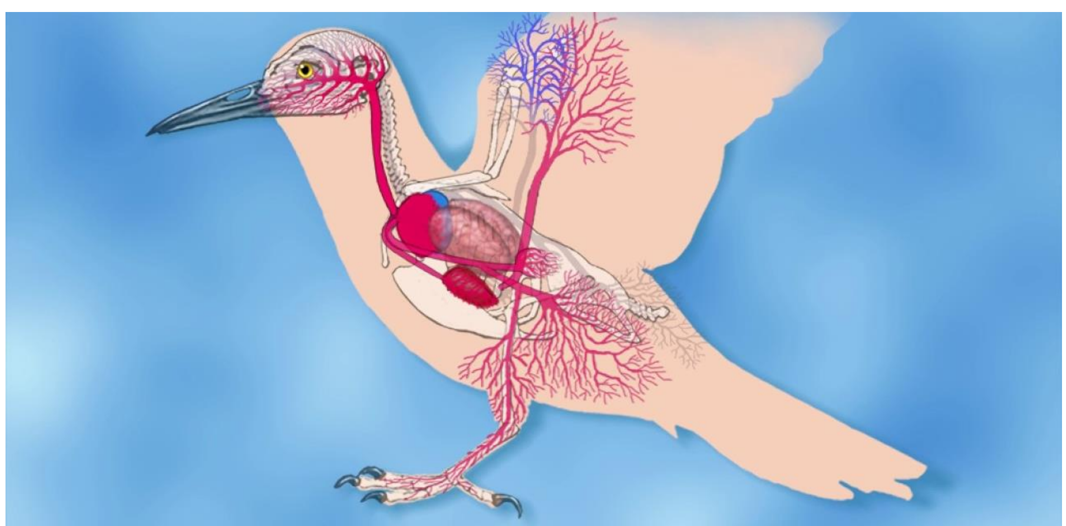




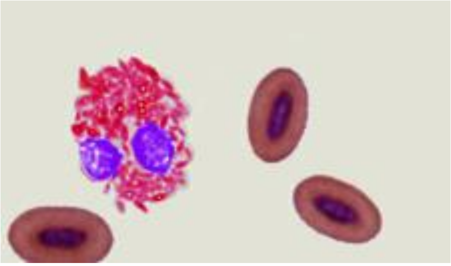
Oxygenated blood (red) is pumped from the lungs to various parts of the body; depleted blood (blue) is returned for recycling.



Systemic and pulmonary circulations are served by one half of a dual heart.

Bird red blood cells (erythrocytes) are nucleated and biconvex. RBC of birds contains large amount of haemoglobin and this is responsible for blood's quick and perfect aeration.

Mobile phagocytes are active and efficient in repairing wounds and destroying microbes



Elliptical in shape and nucleated . Life span 25 - 45 days

Many bird bones are pneumatic (penetrated by air sacs) and do not contain marrow.

Hemopoietic bone marrow (red-blood-cell-producing marrow) is located in the radius, ulna, femur, tibiotarsus, scapula, furcula (clavicles), pubis,and caudal vertebrae.

Respiration

The bird respiratory system differs radically from the lungs of both reptiles and mammals.

As a flying bird requires great and sustained power, therefore, its respiratory system remains specialized to meet the extensive rate of metabolism greater amount of oxygen molecules is needed by the body tissues.

For this purpose, the dense, inelastic and complicated lungs are supplemented by a remarkable system of air sacs, which grow out from lungs and occupy all available space between internal organs, even extending to the cavities of hollow bones.

The air sacs primarily reduce the specific gravity of the bird and also facilitate complete aeration of the lungs.

The avian lungs are aerated twice at each breath which is called double respiration.

The air sacs help in regulating body temperature by internal perspiration.

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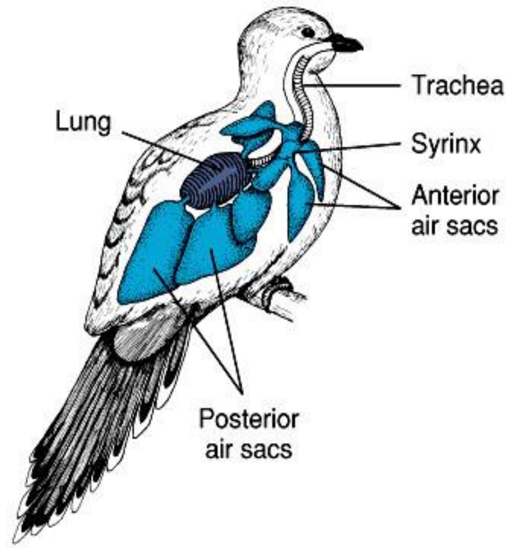
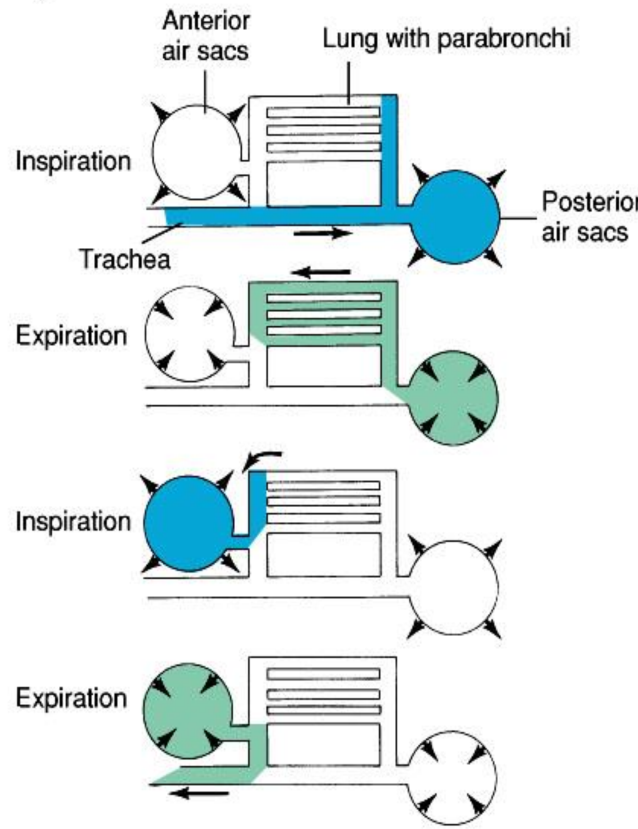
Respiration

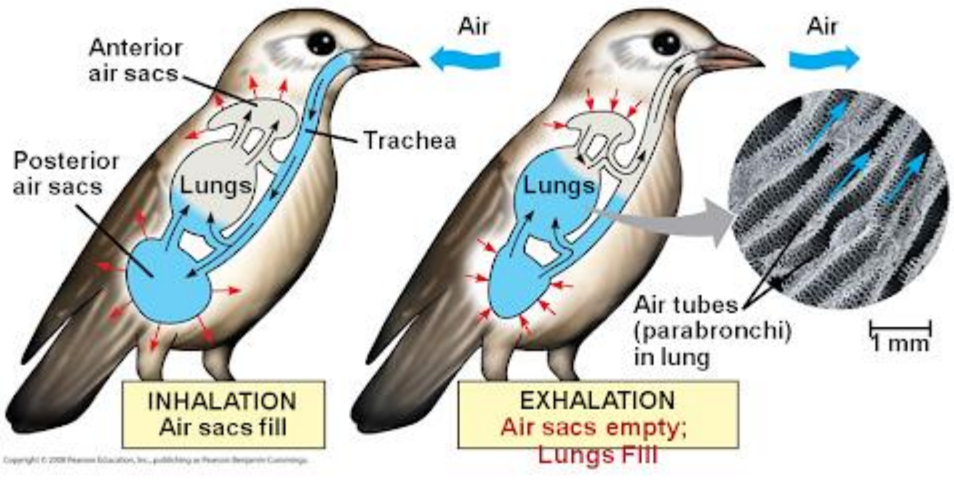
- Bird Lungs
  - The finest branches of the bronchi do not terminate in alveoli but are tube-like parabronchi.
  - Air sacs extend into the thorax, abdomen, and even the long bones.
  - A large portion of the air bypasses the lungs and flows directly to the air sacs on inspiration.
  - On expiration, this oxygenated air flows through the lungs; therefore there is continuous air flow.





Respiration

- Thus it takes two respiratory cycles for a single breath of air to pass through the system.
- This is the most efficient respiratory system of any vertebrate.
- An air sac system helps cool a bird during vigorous exercise when up to 27 times more heat is produced.
- The air sacs extend into bones, legs and wings, providing considerable buoyancy to the bird.



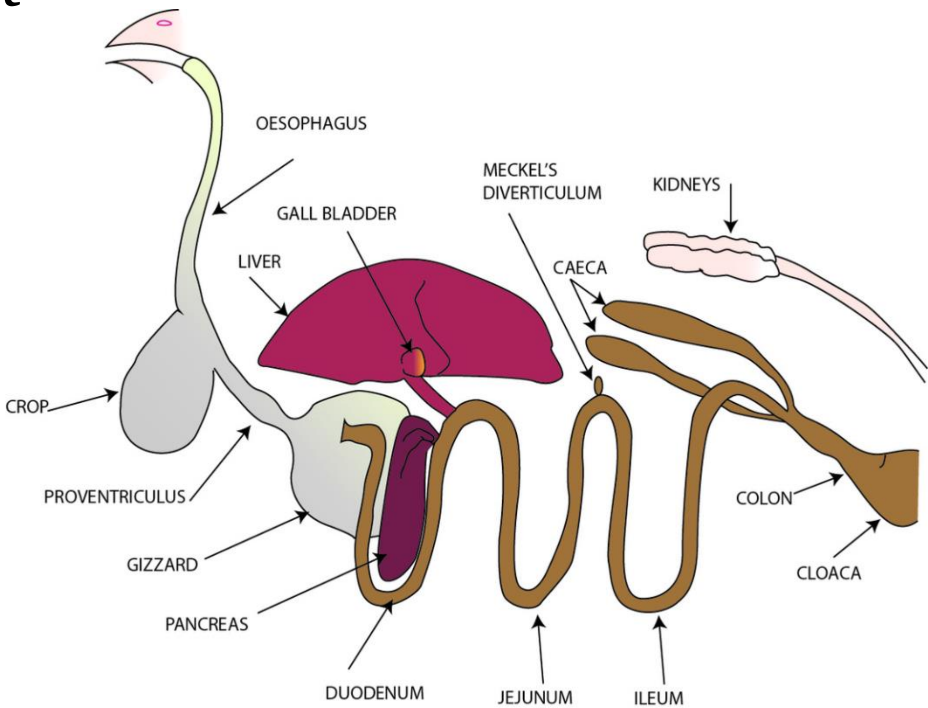
Warm-bloodedness :

In birds the body temperature remains high (40°-46°C) and does not change with the change of environmental temperature.

The high and constant body temperature enables the bird to take flights at high altitudes and also facilitates activeness in every season.

Excretory System

- Birds also use the reptilian adaptation of excreting nitrogenous wastes as uric acid.
- In shelled eggs, all excretory products remain within the eggshell; uric acid is stored harmlessly.
- A bird kidney is less efficient than a mammal kidney in removing ions of sodium, etc.



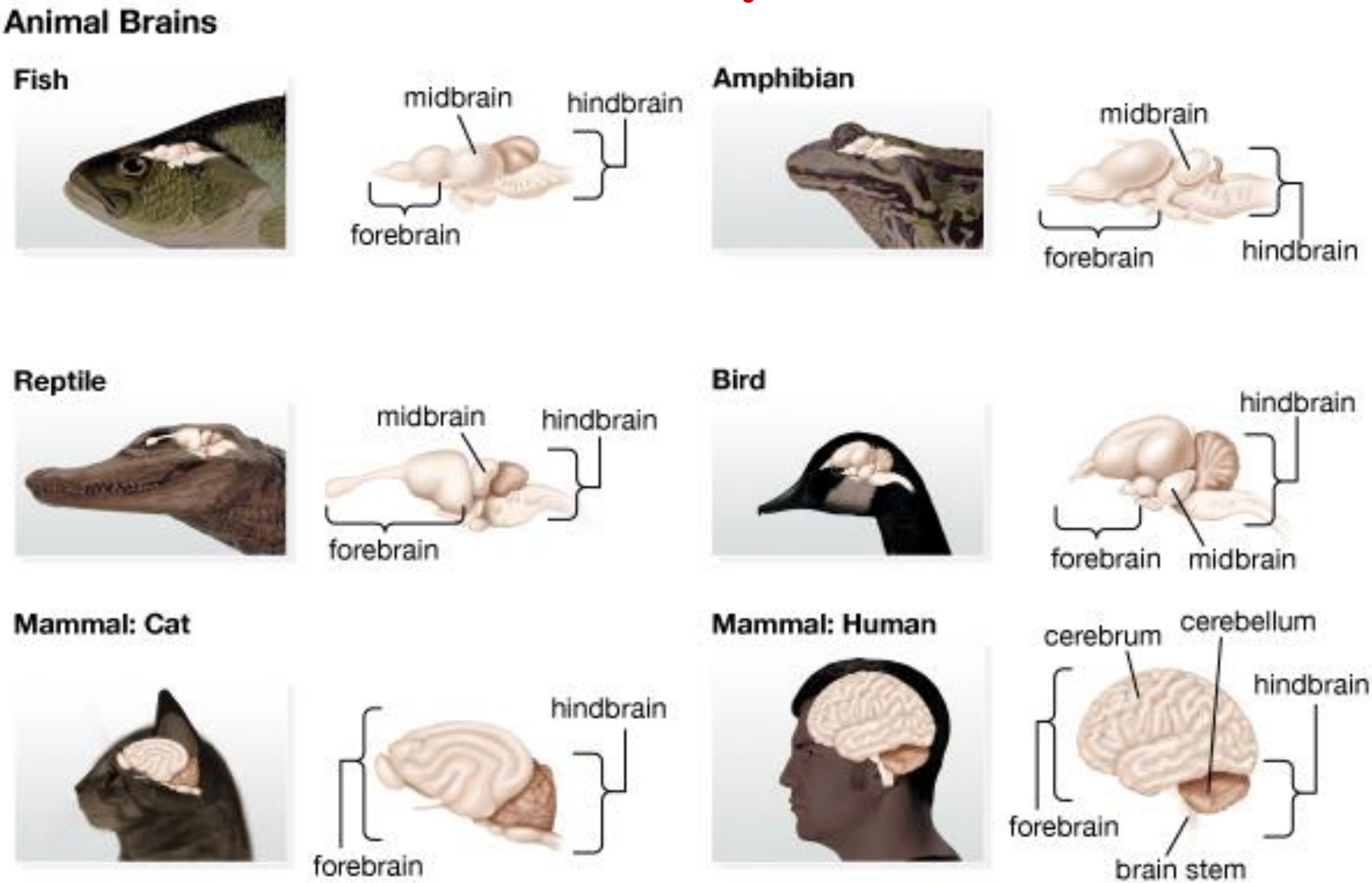
The avian excretory system becomes specialized in three ways

1. For the retention of water, the uriniferous tubules of avian metanephric kidneys are added with Henle's loops, which are efficient in water absorption. coprodaeum of cloaca is another efficient water absorbing organ of birds.

2. For reducing the weight of body, there occurs no urinary bladder and the semi-solid urine is immediately excreted out, not retained for long in the body.

3. The metabolic nitrogenous wastes are converted into less toxic and insoluble organic compounds such as uric acid and urates, which is an important physiological volant adaptation.

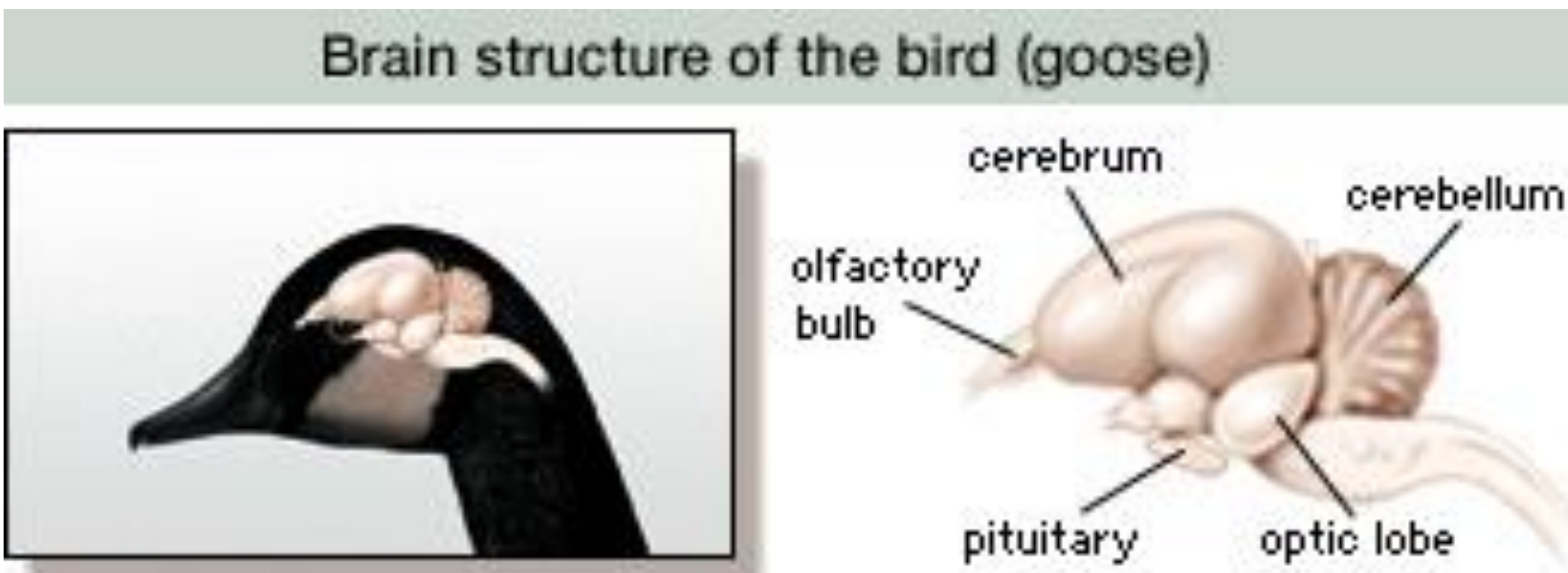
Nervous System



Nervous System

Brain Structures that help birds

The forebrain of birds is much larger than that of reptiles due to the enlargement of the cerebral hemispheres including a region of gray matter, the corpus striatum (functions in visual learning, feeding, courtship, and nesting)

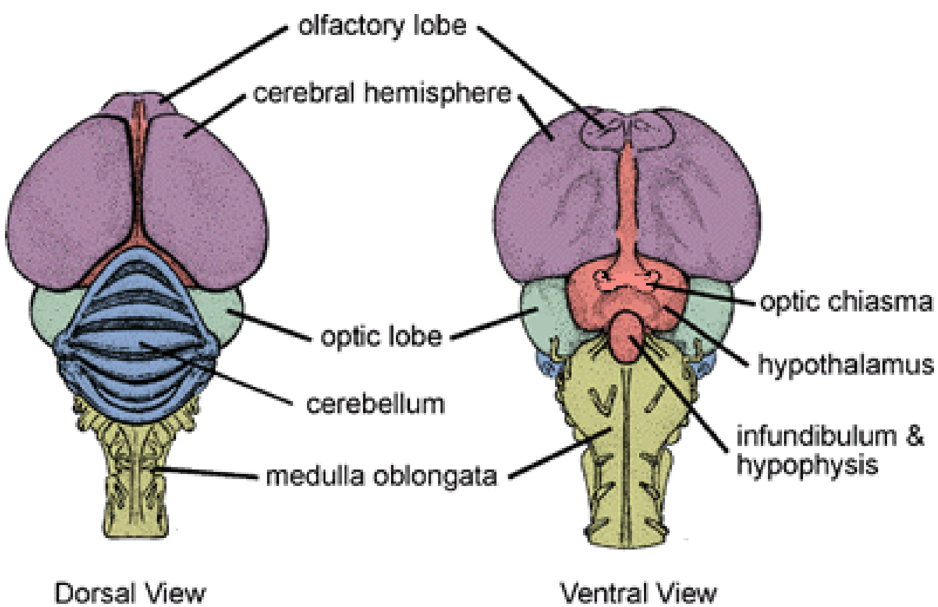


Pineal body – stimulates ovarian development and regulates functions influenced by light and dark periods

Optic tectum – integrates sensory functions

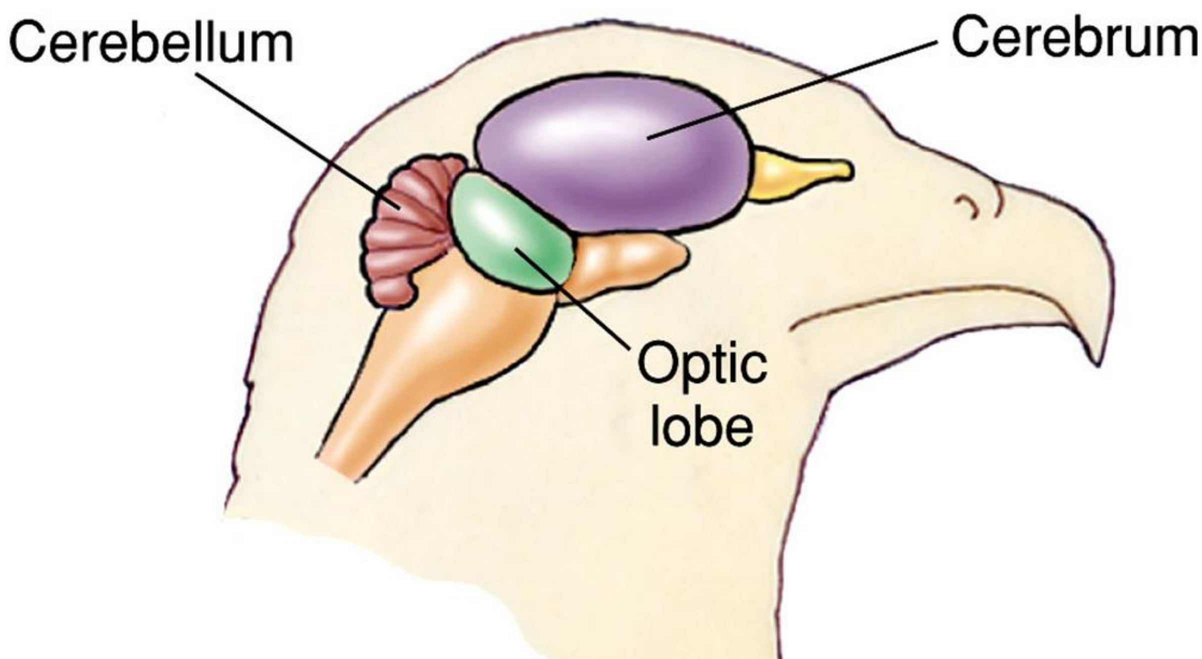
Midbrain – sensory inputs from eyes

Hindbrain – coordinates motor activities and regulates heart and respiratory rates



The avian brain is highly developed consisting of well-developed centers of equilibrium, muscular coordination and instinct. Hence, the cerebellum and cerebrum are highly developed.

Because birds have to depend mostly on the sense of sight, so, the eyes and optic lobes of brain are well developed. Presence of pecten in the eye for assisting vision.



Special Senses

- The optic lobes bulge to each side of the midbrain and form a visual association apparatus.
- Sense of smell is poorly developed except in flightless birds, ducks and vultures.
- Birds have good hearing and superb vision, the best in the animal kingdom.
- The bird ear is similar to the ear of mammals.
  - The external ear canal leads to an eardrum.
  - The middle ear contains a rod-like columella that transmits vibrations to the inner ear.
  - An inner ear has a short cochlea; it allows birds to hear about the same range of sound as humans.

Vision

- Eye is similar to the mammal eye, but it is relatively larger for a given body size
- A bird eye is less spherical and almost immobile; a bird turns its head rather than its eyes.
- The light-sensitive retina has both rods and cones.
- Diurnal birds have more cones; nocturnal birds have more rods.

Like Reptiles, Birds have a nictitating membrane that is drawn over the surface of the eye to cleanse and protect the eye

Eye Placement

- Vegetarians must avoid predators and they have eyes placed to each side to view all directions.
- Birds of prey have eyes directed forward to provide better depth perception.
- Many birds have two foveae or regions of detailed vision; this provides both sharp monocular and binocular vision.
- A hawk has eight times the visual acuity of a human and can see a rabbit over a kilometer away.
- An owl's ability to see in dim light is more than ten times that of a human.
- Many birds can see partially into the ultraviolet spectrum, seeing flower nectar guides.

Reproductive organs

In female birds the ovary and oviduct of one side (i.e., right side) of body are preserved. This necessarily reduces the weight of body.

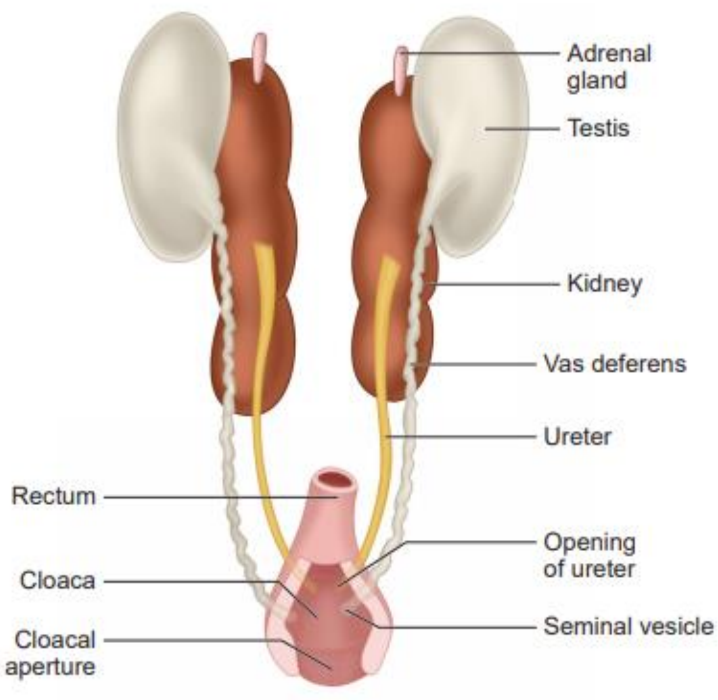


Figure 4.36 Pigeon –Male Urinogenital System

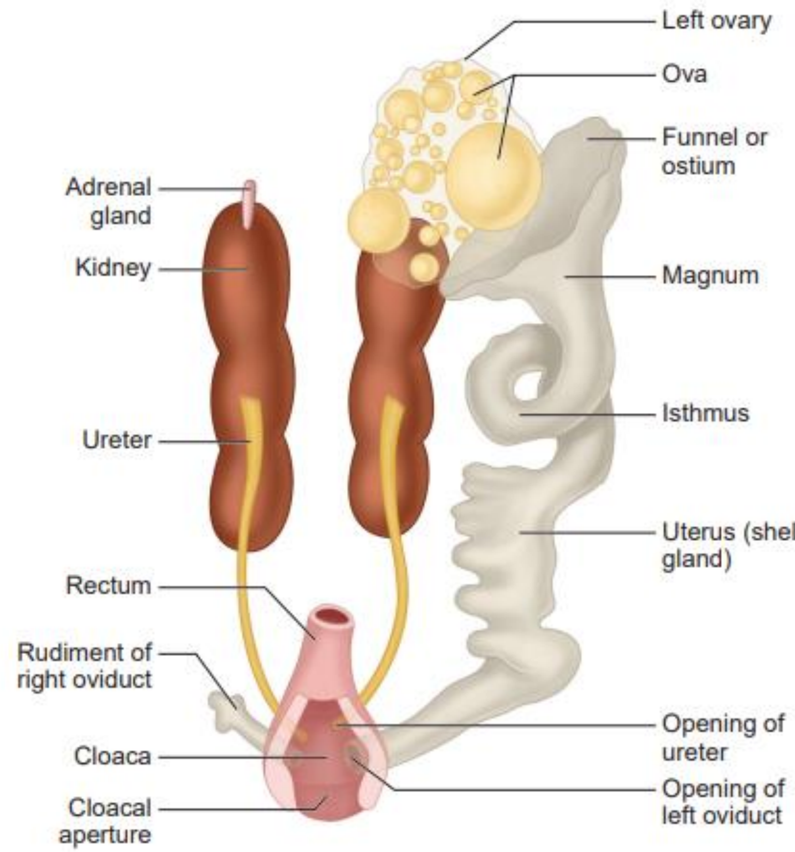
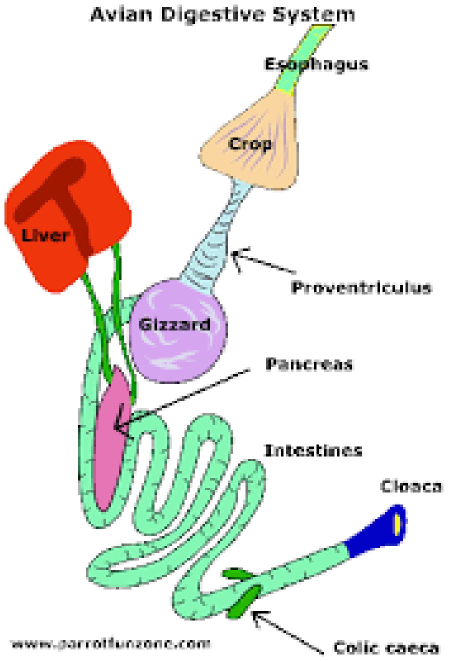


Figure 4.37 Pigeon –Female Urinogenital System

Digestive system

The rate of metabolism in birds is very high, food requirements are great and digestion is rapid.

Most birds are very selective in their diet and accordingly their beaks are variously modified. Further, because undigested waste is minimum and is immediately got rid off, consequently the rectum becomes much reduced in length and never stores the undigested food.



The ill-development of rectum of flying birds indicates towards the fact that the flying animals cannot afford to bear the weight of faeces.

The absence of Gall bladder to reduce weight.

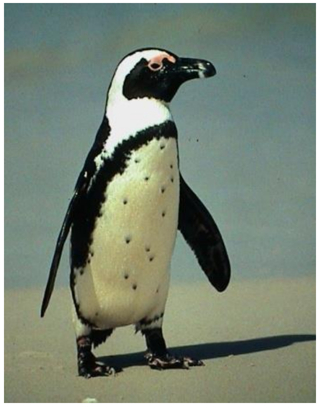
Avian Anatomical Adaptations for Flight:

???

1) Streamlined body:

Reduced resistance when moving through air…

vs.



2) Feathers:

(a).The smooth closely fitting and backwardly directed contour feathers make the body stream-lined and reduces the friction

(b) The feathery covering makes the body light and at the same time protects from the hazards of environmental temperature,

(c) The feathers hold a considerable blanket of enveloping air around the body and add much to its buoyancy.

(d) The non-conducting covering of feathers insulates the body perfectly and prevents loss of heat to maintain a constant temperature.

Avian Anatomical Adaptations for Flight:

3) Bones:

A) Pneumatic: (air-filled; reinforced with internal struts (= trabeculae)

- Reduced weight (but see diving birds…)

B) Number reduction: (weight)

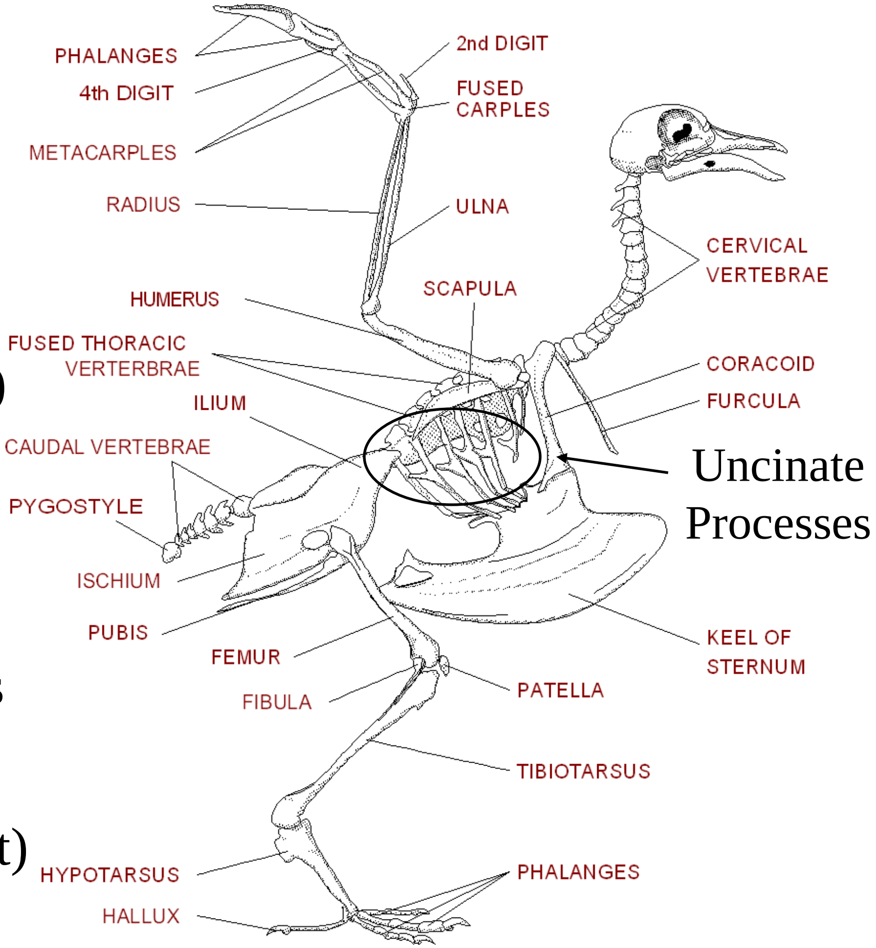
- No teeth
- Carpal / tarsal bone reduction
- Digits lost

C) Fusion: (strength / stability)

- Thoracic vertebrae fused (platform)
- Synsacrum (pelvic support)
- Pygostyle (tail feather support)
- Furculum (wishbone – "spring")
- Carpometacarpus / Tarsometatarsus

D) Additional Modifications

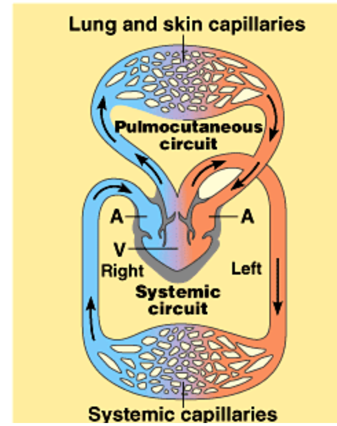
- Keeled sternum (muscle attachment)
- Enlarged humerus (major force…)

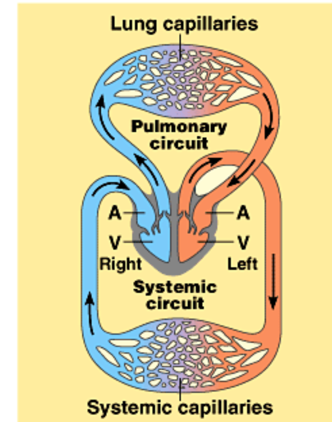


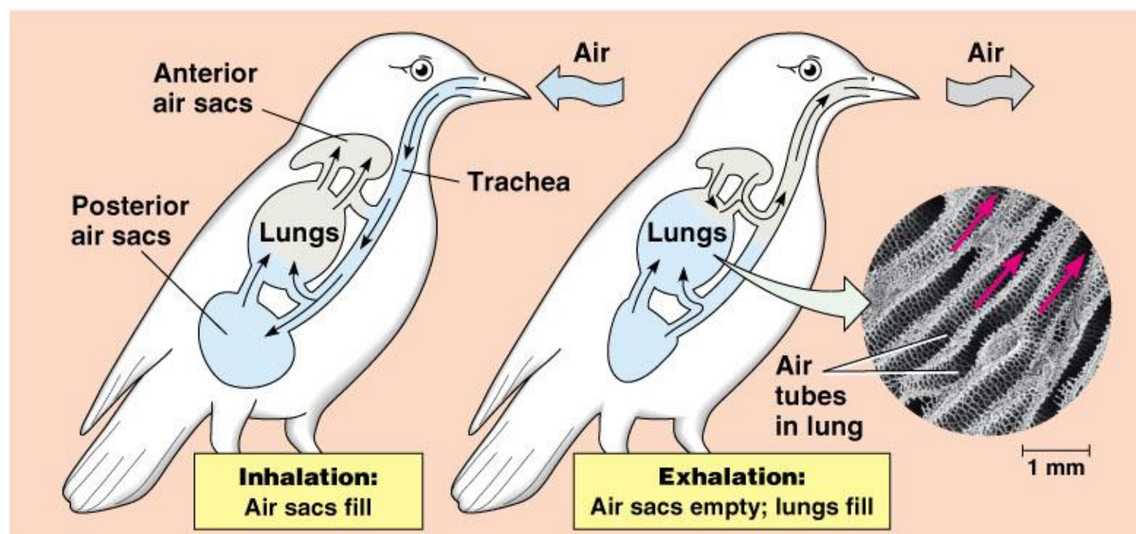
Flight:

Required Modifications for Endothermy:

1) Cardiovascular System
    - Larger, more muscular heart
        - $\uparrow$ blood flow / $\uparrow$ blood pressure
        - Separation of O_2-rich / O_2-poor blood
    - $\uparrow$ hemoglobin concentration in blood
2) Respiratory System
    - $\uparrow$ exchange surface / unit lung volume
    - Unidirectional air flow (no mixing of fresh / stale air)
    - $\uparrow$ blood flow to lungs
3) Insulation (= feathers)







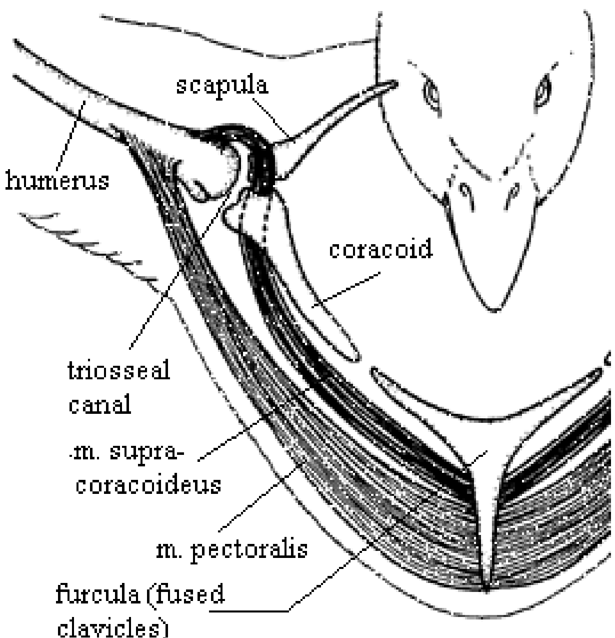
Avian Anatomical Adaptations for Flight:

4) Muscles:

- Muscle reduction
  - Jaws – power not necessary (food swallowed whole / in chunks)
  - Legs – rigid skeleton provides support; perching only major requirement
    - Muscle centralized on proximal portion of limb (center of gravity)
- Flight muscles
  - Size increase; location near center of gravity
    - Both up-stroke (supracoracoideus) and down-stroke (pectoralis) muscles originate on keel

5) Forelimbs Modified as Wings:

Airfoils to generate lift…



Flight:

Parachuting
(drop with little control)

Gliding
(membranes = ↑ lift)

Soaring
(utilize wind currents)

Passive Flight
(requires little energy…)

Active Flight
(requires lots of energy…)

Evolution of Active Flight:

"From the Tree Down" Theory of Flight:

- Early ancestors tree climbers – jumped from tree to tree
  - Selective pressure favored increased distance / lift

Gliding → Weak Flapping → Full Airborne Flapping

"Arborealists"
(ornithologists)

"From the Ground Up" Theory of Flight:

- Early ancestors fast bipedal runners – "wings" lightened load
  - Flapping evolved to provide additional forward propulsion
- Early ancestors used "wings" to snare insects
  - Flapping evolved to assist horizontal jumps for prey
- Early ancestors used wings to run up steep slopes
  - Wing Assisted Incline Running

"Cursorialists"
(paleontologists)

Flight:

Cost / Benefit of Flying:

1% of energy expended per mile covered (versus mouse)

Costs:

- Energetically costly (short-term)
- Limits range of body size

Benefits:

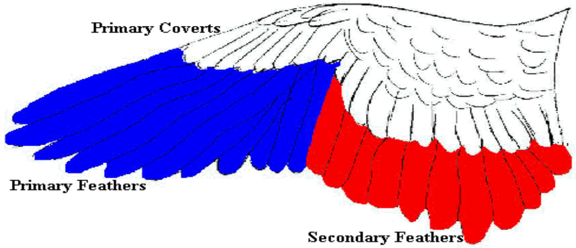
- ↓ cost / unit distance
- Exploitation of new resources
- Escape from predators

Metabolic / Energetic Requirements for Flight:

Bird's core temperature higher than similar sized mammal

- Reduction in Weight (= reduce cost of flight)
  - Visceral organs small but efficient; pneumatic skeleton
- High Metabolic Rate (↑ ability to sustain muscle activity)
  - **Endothermy**: Core temp. sustained by heat released from metabolic processes
    - Advantages:
      1) Faster response time for brain / muscles
      2) Activity levels maintained in cold environments
    - Disadvantage: ↑ caloric intake required

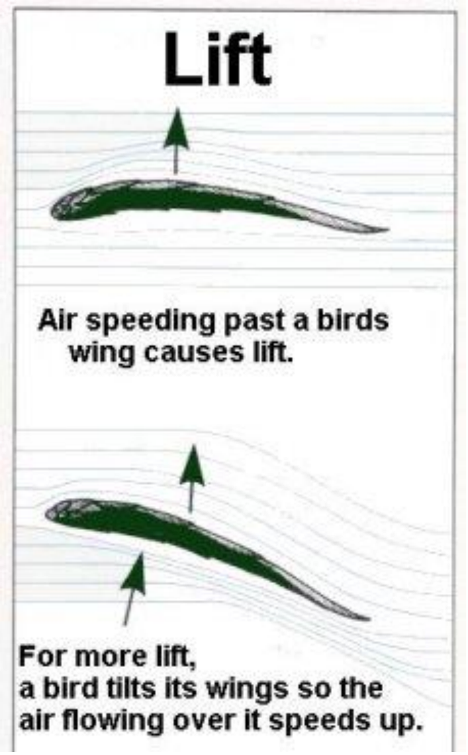
Mechanics of Wing Design:

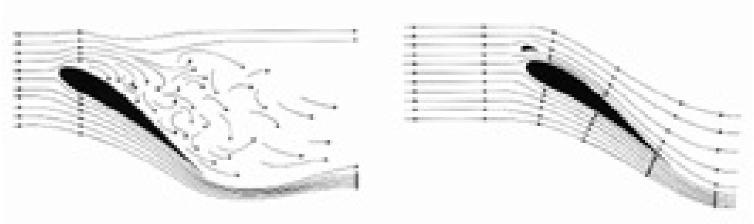


During upstroke, air passes between flight feathers = cost reduction…

- Two types of contour feathers present on wing:
    1) Primaries: Located on hand; provide thrust
    2) Secondary: Located on back of arm; provide lift
- Force produced as air passes wing:
    1) Lift = Vertical force equal to or greater than weight of bird
        A) Cambered Airfoil
            - Upward curvature of wing; tapers toward back
            - Ventral = High pressure; Dorsal = Low pressure
            - ↑ camber allows for flight at slower speeds
        B) Angle of Attack
            - Leading edge of wing tilted; ↓ pressure dorsally
            - Stalling angle = airflow separates from wing (~ 15º)
            - Alula "Bastard wing"
                - Reduce drag by improving air flow over wing (= steeper angle of attacks)

Bernoulli principle

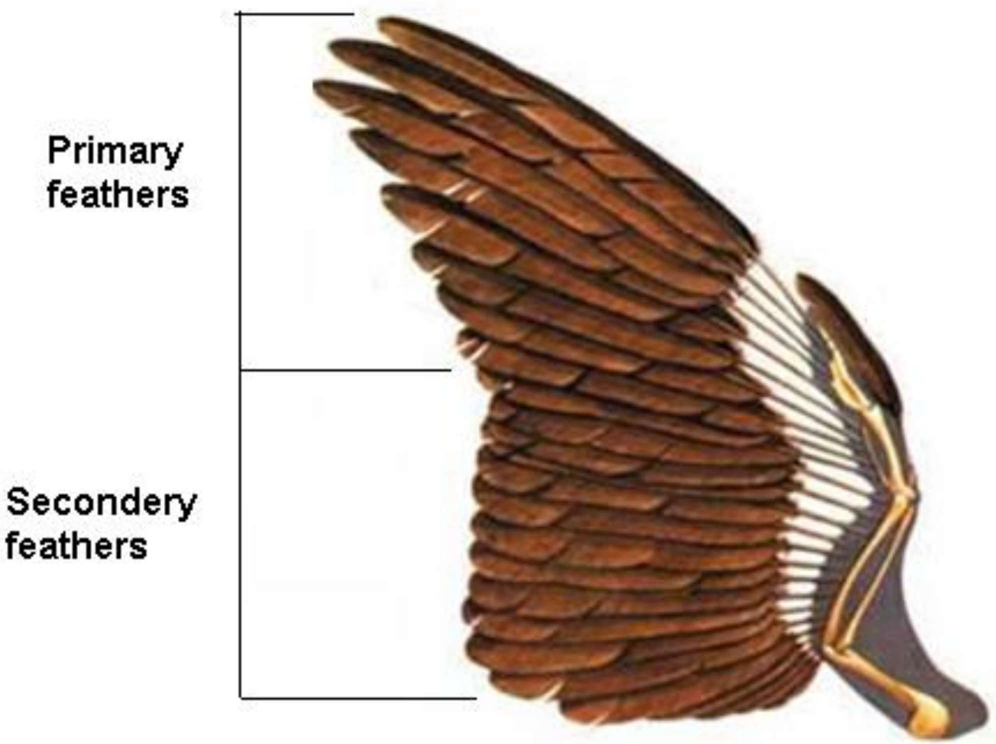


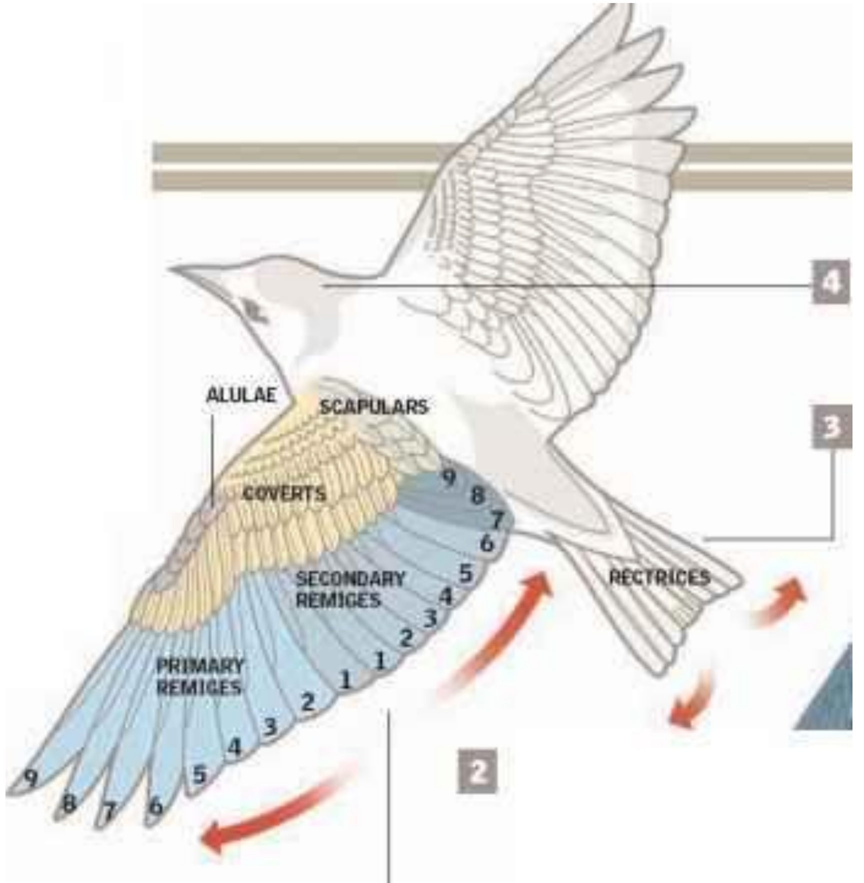


Organs for flight:

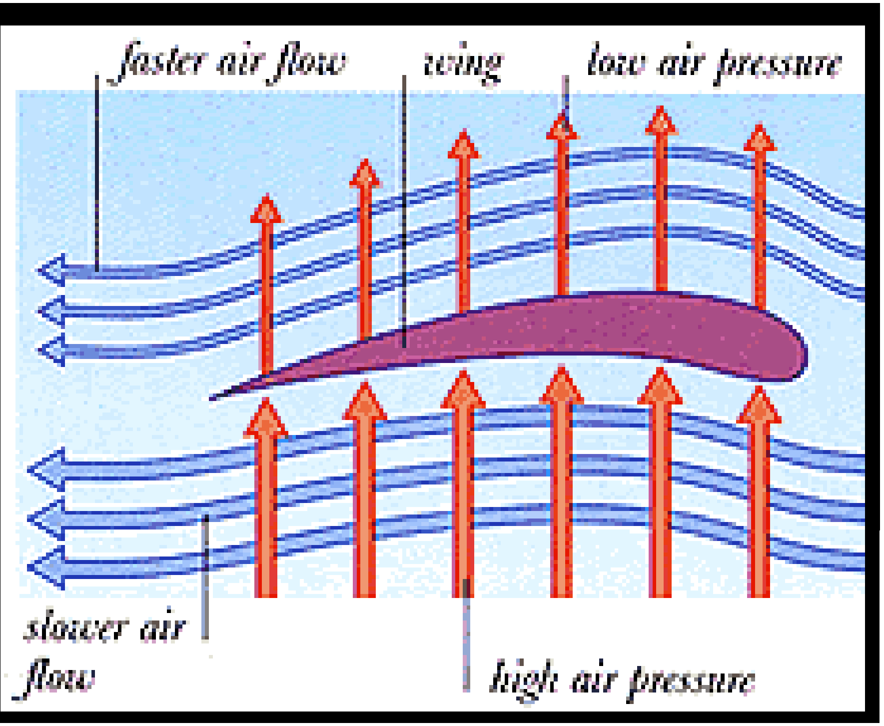
The forelimbs have transformed into unique and powerful propelling organs, the wings. The wings are the sole organs of flight.

Feathers are absolutely adapted for the flight. These structures arose their thermoregulatory role and made possible subsequent evolution of flight.

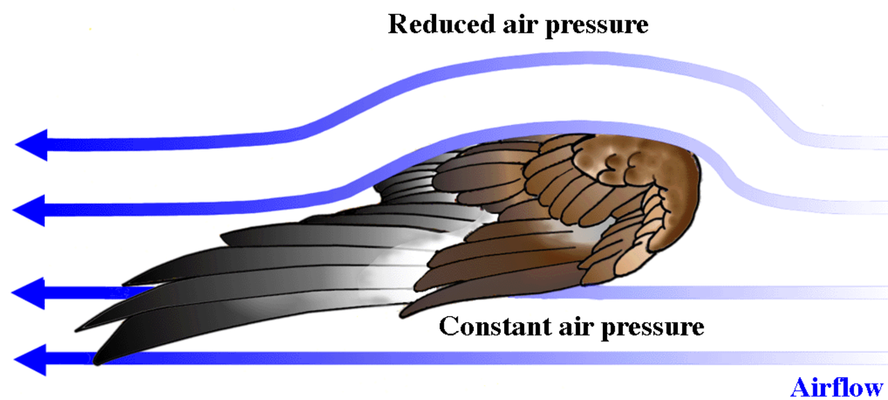


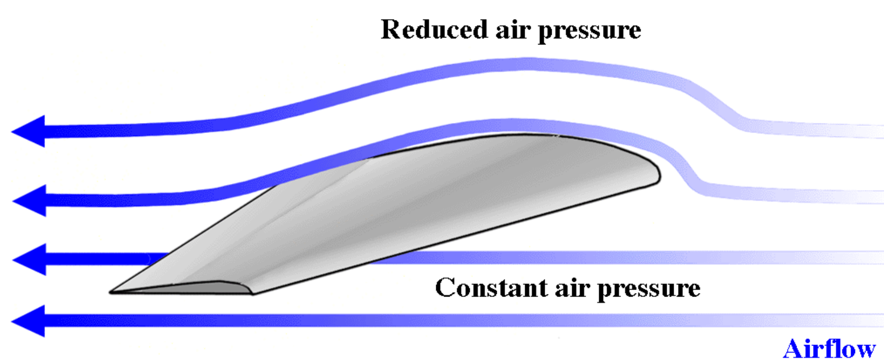


- Mechanics of Bird Flight:
  - Increasing the angle of attack increases lift but also increases turbulence.
  - The alula reduces turbulence.
  - The alula is a group of small feathers that bones of the medial digit support.
    - During take off, landing, and hovering flight, the angle of attack increases the alula is elevated
    - During soaring the fast flight, the angle of attack decreases and slotting is reduced.

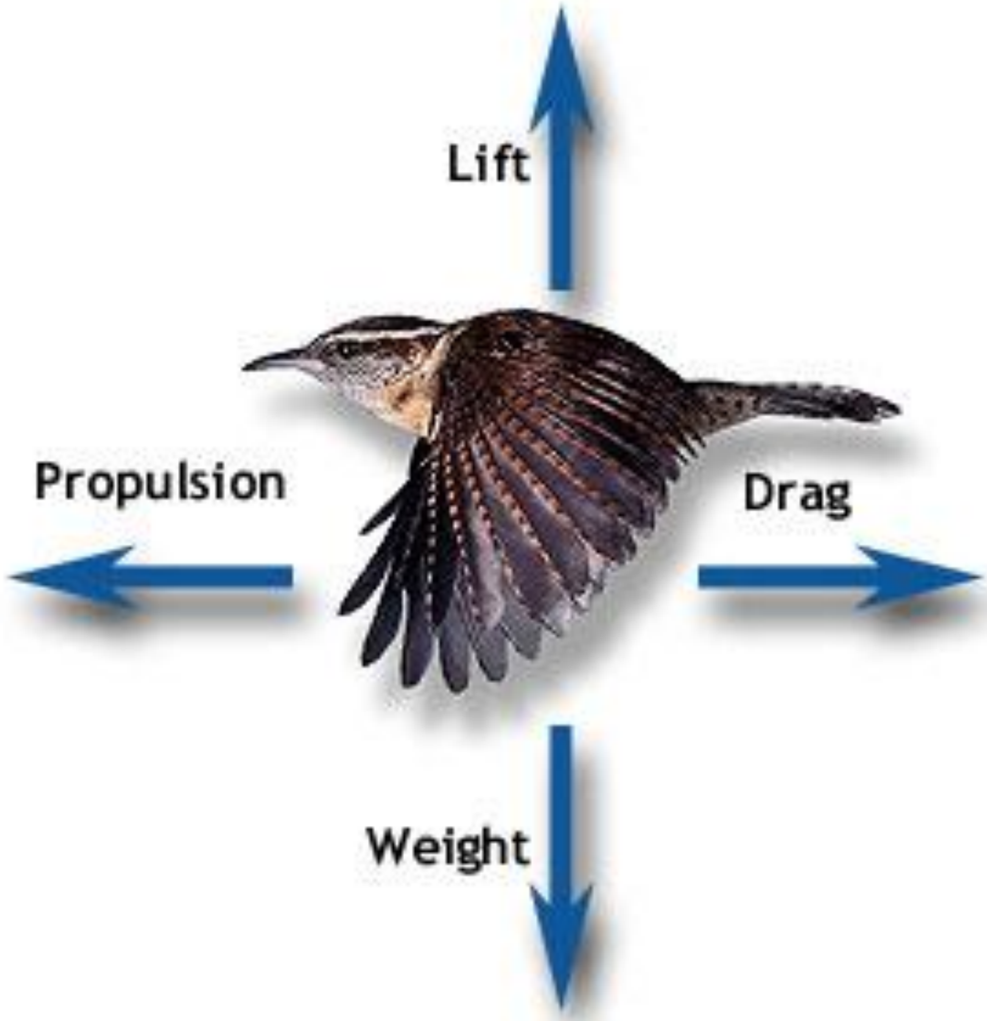


This makes low pressure over the top and higher pressure over the bottom. The wind pulls the wing up from the top and pushes it up from the bottom. It creates "lift". Airplane wings work the same way.





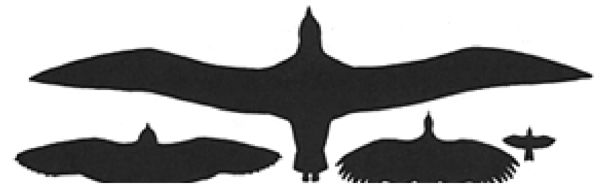
http://www.paulnoll.com/Oregon/Birds/flight-lift.jpg


Lift
Propulsion
Drag
Weight

Flight:

Mechanics of Wing Design:

- Force produced as air passes wing:
  2) **Drag** = Component opposing forward movement; created by turbulent air flow



  - Highest at tips of wings
  - Reduce effect = 1) Taper wing
    2) Lengthen wing

  - Aspect ratio = measures amount of wing drag produced, relative to lift

    Ratio = wing span / wing width

    Low Aspect Ratio = Wide, short wings (higher drag)

    High Aspect Ratio = long, narrow wings (lower drag)

- **Wing Loading**: (Ratio = body mass / wing surface area)
  - Correlates to size – body mass ↑ faster than surface area as body size ↑
    - Low Wing Loading = more maneuverable; ↓ power needed to sustain flight
    - High Wing Loading = less maneuverable; often soaring birds

Basic Forms of Bird Wings

- **Elliptical Wings**

1) Birds that must maneuver in forested habitats have elliptical wings.
2) Elliptical wings are slotted between primary feathers to prevent stalling at low speeds, etc.
3) The small chickadee can change its course within 0.03 seconds.

- **High-Speed Wings**

1) Birds that feed on the wing or make long migrations have high-speed wings.
2) These wings sweep back and taper to a slender tip; this reduces "tip vortex" turbulence.
3) They are flat in section and lack wing-tip slotting.

Basic Forms of Bird Wings

- **Soaring Wings**

1) Albatrosses, gannets and other oceanic soaring birds have wings with long, narrow wings.

2) The high-aspect ratio of long, narrow wings lack wing slots and allow high speed, high lift and dynamic soaring.

3) They have the highest aerodynamic efficiency of any design, but are less maneuverable.

4) These birds exploit the highly reliable sea winds and air currents of different velocities.

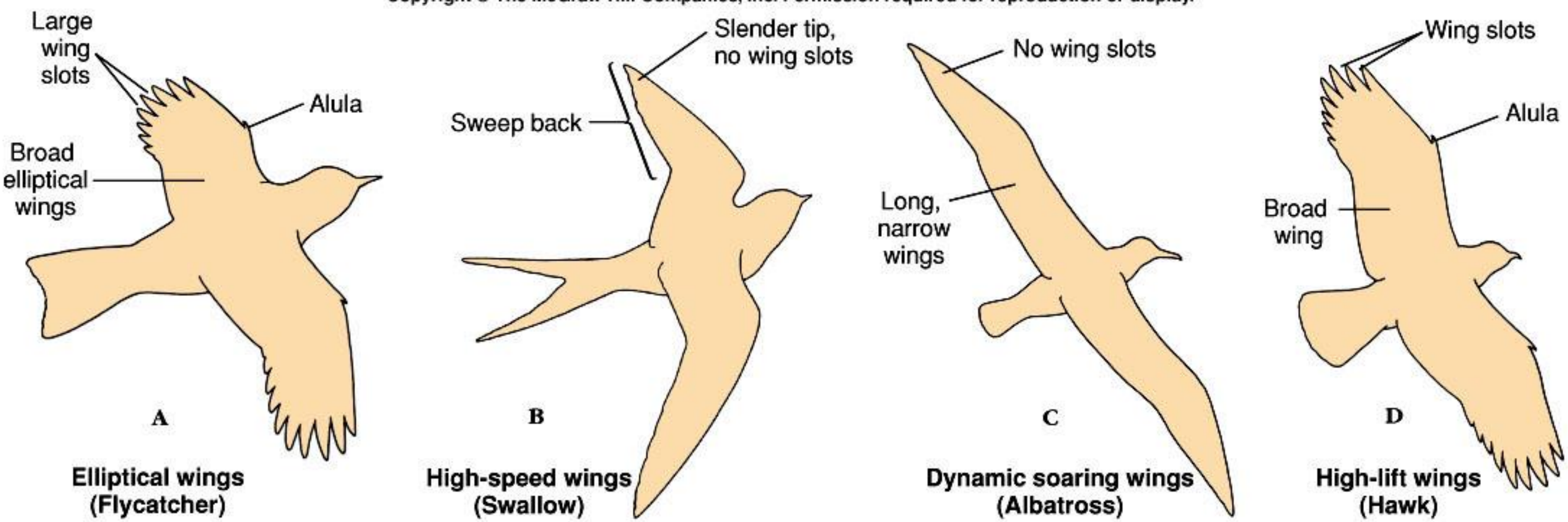
Basic Forms of Bird Wings

High-Lift Wings

1) Vultures, hawks, eagles, owls and other birds of prey that carry heavy loads have wings with slotting, alulas and pronounced camber.

2) This produces high lift at slow speed.

3) Many are land soarers; their broad, slotted wings allow sensitive response for static soaring.



Elliptical wings (Flycatcher)

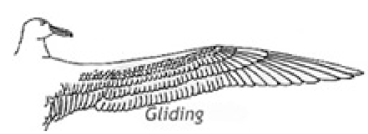
High-speed wings (Swallow)

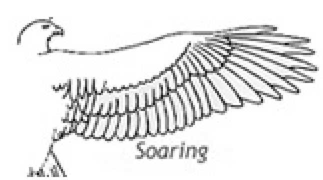
Dynamic soaring wings (Albatross)

High-lift wings (Hawk)

Major Structural Wing Types:

Characteristic	Elliptical (short & rounded)	High Speed (Taper to point)	Dynamic Soaring (Long & narrow)	High Lift (Long & broad)
Examples:	Sparrow Robin Pigeon	Swallow Duck Falcon	albatross shearwater petrel	eagle vulture raven
Camber	High	Low	Very low	High
Alula	Large	Absent	Absent	Large
Speed	Slow	Fast	Fast	Slow – Mod.
Acceleration	Fast	Slow	Slow	Fast
Maneuverability	High	Moderate	Very Low	High
Endurance	Low	High	Very High	Moderate
Aspect Ratio	Low (3 – 6)	Moderate (5 – 9)	Very high (9 – 18)	Moderate (6 – 7)
Wing Loading	Low (small birds)	Low (small birds) High (large birds)	Moderate - High	Moderate (can pick up weight)





Avian Anatomical Adaptations for Flight:

3) Bones:

A) Pneumatic: (air-filled; reinforced with internal struts (= trabeculae)

- Reduced weight (but see diving birds…)

B) Number reduction: (weight)

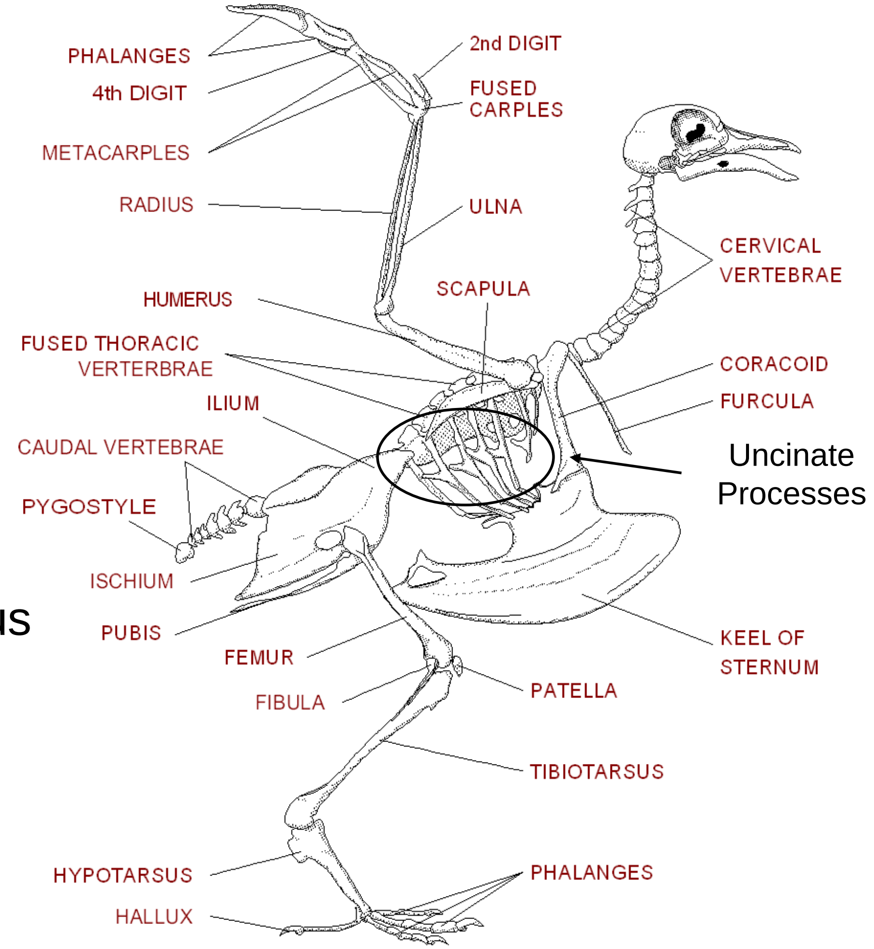
- No teeth
- Carpal / tarsal bone reduction
- Digits lost

C) Fusion: (strength / stability)

- Thoracic vertebrae fused (platform)
- Synsacrum (pelvic support)
- Pygostyle (tail feather support)
- Furculum (wishbone – “spring”)
- Carpometacarpus / Tarsometatarsus

D) Additional Modifications

- Keeled sternum (muscle attachment)
- Enlarged humerus (major force…)



Reproduction

- Reproduce through **internal fertilization**
- During mating male birds press their **cloaca** to the female's **cloaca** and releases sperm
- Produces **amniotic eggs**
  - May lay one egg or a clutch of several eggs

Reproduction and Development

Reproductive Activities of Birds

- Establishing Territories
- Finding Mates
- Constructing Nests
- Incubating Eggs
- Feeding young



Reproduction:

- Utilize colors, postures, and vocalizations for species, sex, and individual identification
  - **Bird Song**: Complex array of notes, often with frequency modulation

    - Often male specific and during breeding season
    - Learned behavior; “window” of opportunity during development
    - Species specific; regional dialects relatively common
    - Function:
      1) Attract mates
      2) Repel rivals
  - Visual displays often associated with song:
    - ♀♀ select males based on visual characteristics :
      - Good nutritional status
      - Low parasite loads
      - Predator avoidance

“truth in advertising…”

♀♀ prefer ♂♂ with longer tails and more eye spots

Offspring grew faster; ↑ survival rates

Reproduction:

- Birds exhibit two broad categories of mating systems:

  - Better quality mate
  - Increased heterozygosity
  - "hedging" your bets

1) Monogamy: Pair bond between single male and female (~ 90% of bird species)

- Both parents required to raise young (e.g., food acquisition)
- Resource distribution (even – control impossible…)
- Skewed sex ratio (partner becomes prized "resource")

  - ➢ Monogamy does not necessarily mean FIDELITY
    - Extra-pair copulation common

Swans, Geese and Eagles mate for life

2) Polygamy:

Individuals mate with > one partner during single breeding season

Polygyny = Single ♂ – multiple ♀♀;

Polyandry = Single ♀ / many ♂♂

Males mate with more than one female, female care for young

Occurs when resources are patchy

Prairie Chickens

Females mate with more than one male

Females are larger and defend territory

Males build nest and care for eggs

Sandpipers

Resource defense polygyny

Male dominance polygyny

Lek: Aggregation of many males

Resource defense polyandry

- All birds are oviparous

Why?

Constraints based on flight

- Most likely ancestral reproductive mode
  - No pressure to evolve vivipary(e.g. endothermy = incubated eggs always warm)

- Genetic sex determination
  - Heterogametic chromosomes – ♀♀ = WZ / ♂♂ = ZZ

Shallow depression

(e.g., Killdeer)

- Nests protects eggs from physical stresses and predation:

Colonial Nesting

Protection in numbers

Egg Incubation

~ 33 – 37 ºC

Megapodes bury eggs

Brood Patch

Feathers lost;

Blood vessels proliferate

Stimulated by prolactin

Elaborate structure

(e.g., osprey)

- Clutch size variable:

1) Trade-off Hypotheses: Driving force is maximization of lifetime reproductive success
    - Physical strain on females / exposure to predation during food collection

2) Predation Hypotheses: Driving force is minimization of nest detection by predators
    - More eggs / young = ↑ detection (sound / smell / trips to nest / etc.)

3) Seasonality Hypotheses: Driving force is food availability during breeding season
    - More eggs / young = ↑ food reserves / ↓ competition

- Young at differing levels of development at hatching:

Precocial	Semiprecocial	Altricial
High yolk	Moderate yolk	Low yolk
Down present	Down present	Down Absent
Eyes open	Eyes open	Eyes closed
Mobile	Semi-mobile	Not mobile
Self-feeding	Not self-feeding	Not self-feeding

Ducks

Hawks

Passerines

Incubation may last from 10 – 80 days

Altricial < Precocial

Growth Rates:

Altricial > Precocial

Infanticide

Nesting and Care of Young

- Nearly all birds lay eggs that must be incubated by one or both parents.
- Eggs of most songbirds require 14 days for hatching; those of ducks and geese may require a month.
- Often the female performs most of the duties of incubation; rarely the male has equal or sole duties.

Parental Care

- Three options for parental care
    1. Only one parent takes care of the egg

       Ex. African Jacana
    2. Caring for hatchling is an affair for the entire family (parents, siblings, …etc).
    3. No care for the young

       Ex. Brown-headed Cowbirds will lay their eggs in other bird's nests and let other birds raise their young



Parental Care

- Most birds care for their eggs and their young
- Birds usually lay their eggs in nests
  - Hold eggs and conceal them from predators and other elements
  - Nests are built from available materials
    - Twigs, barks, grasses, feathers…etc.
- One or both of the parents warm, or **incubate**, the egg by sitting on them.




Bird’s Adaptations

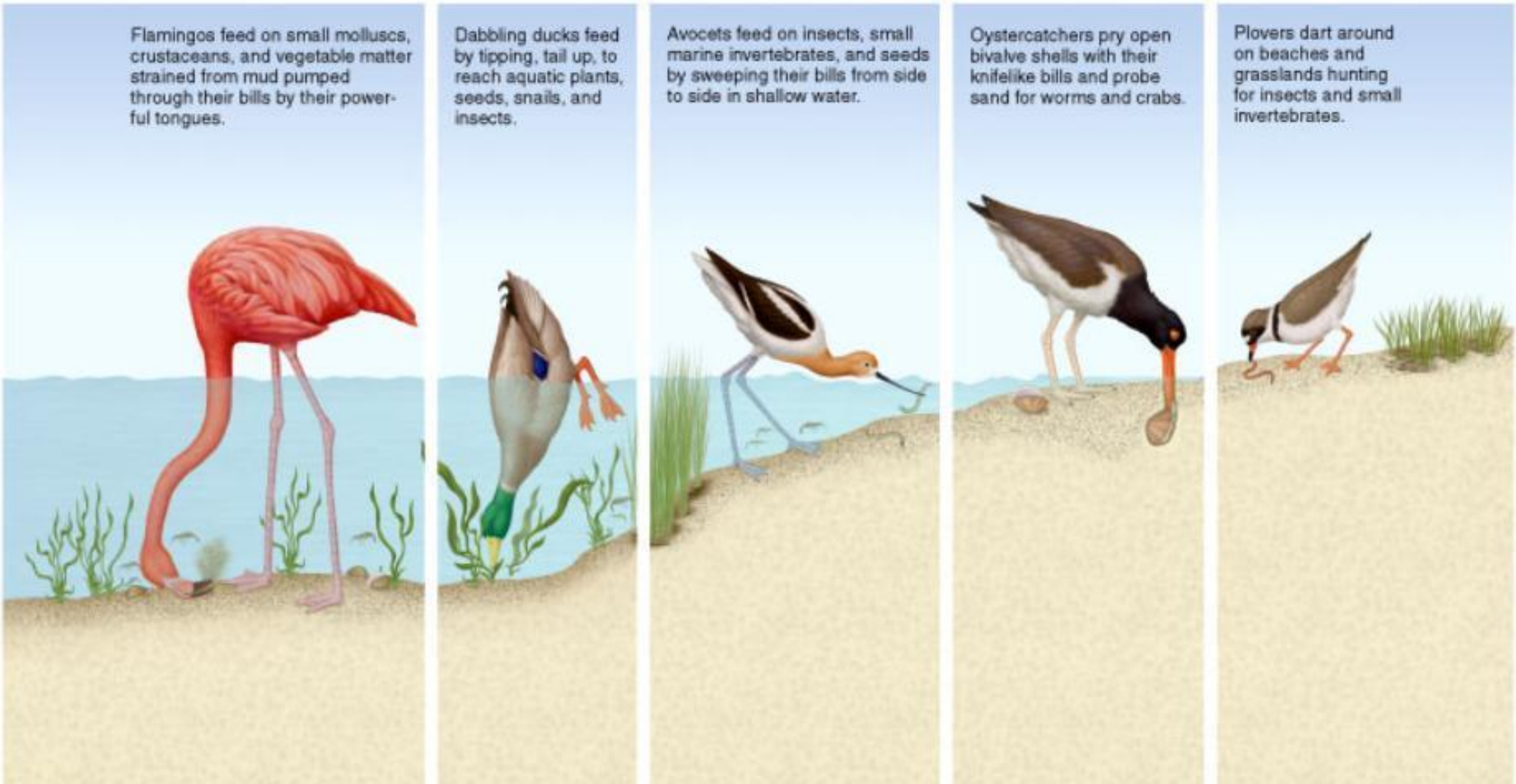
A diverse group, birds exhibit a range of adaptations for all environments and a variety of lifestyles. They can fly, walk, run, swim and dive and occupy the air, oceans, freshwater, seashores, rainforests, deserts and polar regions.

Slowly animals have evolved so that they can survive in their habitats and find food more easily.

Birds have adapted. What a bird’s beak and feet are like depends upon what the bird eats and where it lives

Bird’s beak and feet mainly adapt for the feeding and select their habitats.

Beak Adaptation


Flamingos feed on small molluscs, crustaceans, and vegetable matter strained from mud pumped through their bills by their powerful tongues.
Dabbling ducks feed by tipping, tail up, to reach aquatic plants, seeds, snails, and insects.
Avocets feed on insects, small marine invertebrates, and seeds by sweeping their bills from side to side in shallow water.
Oystercatchers pry open bivalve shells with their knifelike bills and probe sand for worms and crabs.
Plovers dart around on beaches and grasslands hunting for insects and small invertebrates.

Filtering

Ducks have flat beaks with small tooth-like edges to filter plant food from mud and water.

Probing

Ibis, Lipkins, and Black-necked Stilts have probing beaks to poke around in plants and grass to find food.

Catching Insects

Blackbirds like the Crow and song birds have beaks to catch insects.

Cracking Seeds

Some song birds such as the Cardinal have beaks designed for seeds.

Tearing Meat

Birds of prey such as the Osprey have sharp hooked beaks to tear meat.

Mixed

Birds with beaks that are combinations of the above eat a vairety of food.

The Black-crowned Night Heron and Tri-colored Herons here eat mainly fish but will also eat amphibians, crustaceans, leeches, worms and insects.

Foot Adaptation

Grasping: Large curved claws to catch prey.

Scratching: Long nail like toes to scratch in the dirt.

Swimming: Webbed toes to use like paddles for swimming.

Perching: The 4th backward toe is long to reach around branches.

Running: Many fast running birds don't have a backward facing toe and only have 3 forward toes.

Climbing: Many birds who climb the sides of trees have two forward facing toes and two backward facing toes.

Webbed

Birds with webbed feet are swimmers

Lobed

Semi-aquatic Birds like the coot have lobed feet that helps them walk on top of plants in the marsh.

Long and Narrow

These feet belong to the heron with long legs and long toes. They can walk on mucky marsh bottoms.

Short and Narrow

The Ibis has shorter toes and walks on land to look for food.

Perching

The feet shown here are designed for grabbing branches and belong to the parrot.

Feet tell how a bird lives

Marsh wren
perching foot; three toes forward; one long toe backward

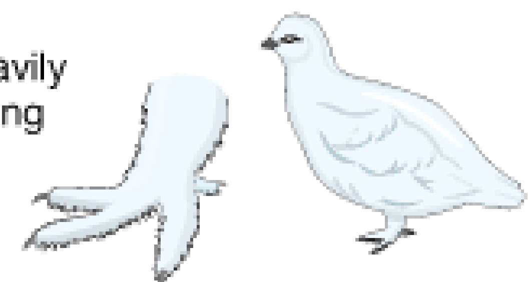
Sapsucker
climbing foot of woodpecker; two toes forward; two backward

Osprey (fish hawk)
catching, holding foot; long, sharp claws (talons)

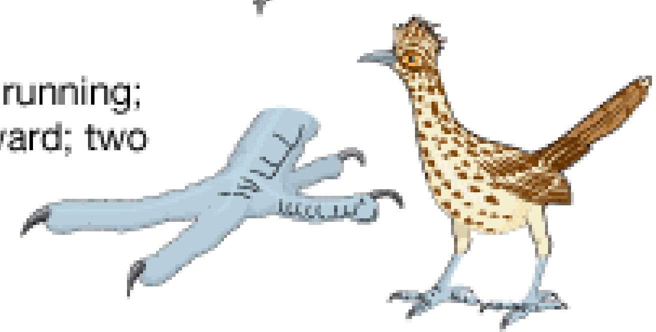
Nighthawk
weak foot; not used to perch; bird crouches on flat surface

Scaup duck
webbed foot, used like a paddle for swimming and diving

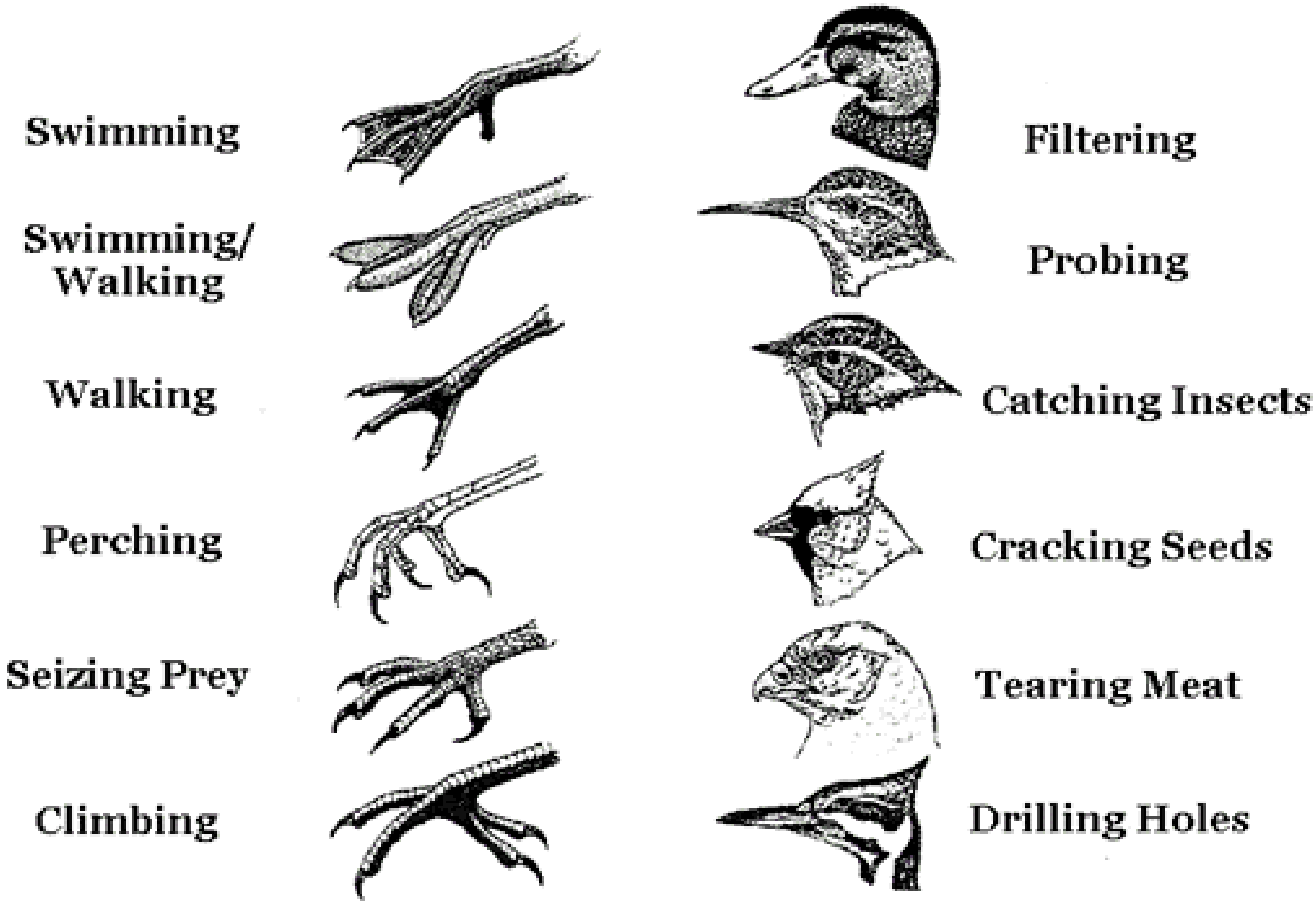
Ptarmigan
snowshoe foot, heavily feathered, for walking on snow



Roadrunner
large foot for running; two toes forward; two backward



Bird Beaks and Feet



Migration and Navigation

- Migration – refers to periodic round trips between breeding and non-breeding areas.

- Migration is the yearly, seasonal journey undertaken by many species of birds. During this journey, birds cover distances of many kilometers.

- Most migrations are annual
  – Nesting regions north and wintering regions south

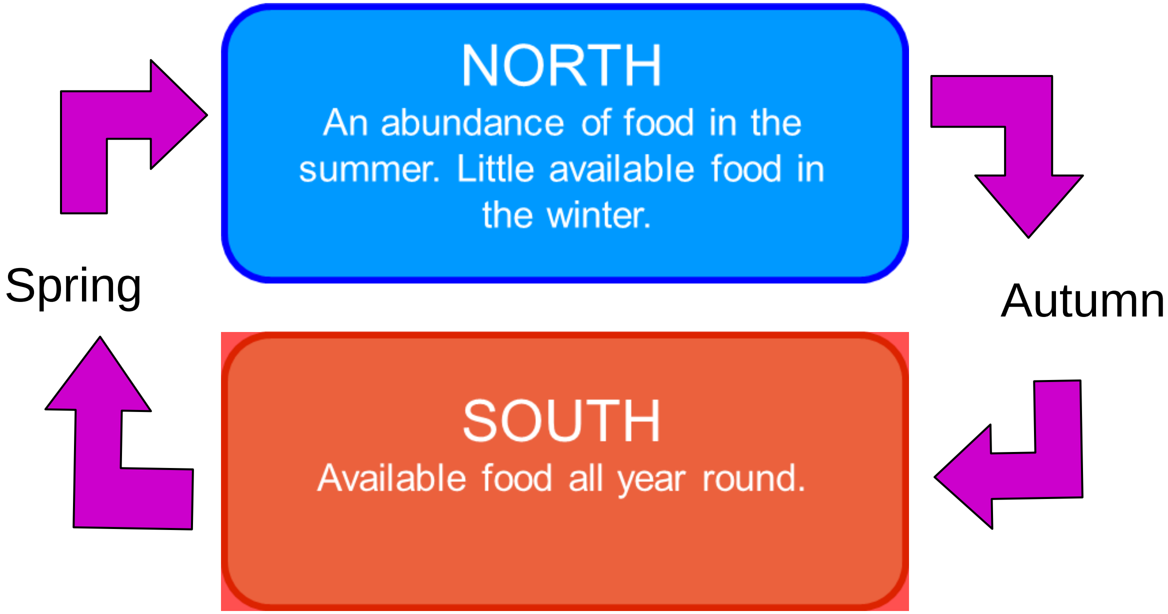
- Migration allows birds to avoid climatic extremes and to secure adequate food, shelter and space throughout the year.

The most common types of migration are those carried out by birds in the spring and the autumn. In the autumn, they travel from breeding grounds in the north to wintering grounds in the south, and vice versa in the spring.

Why do birds migrate?

Birds depend on the amount of available food in an area to survive and raise their chicks, so when there is a shortage of food in one area, they move to areas where more food is available.

Changes in the amount of available food in certain areas are related to changes in temperature. These changes occur in different seasons in different regions of the world.



Migration and Navigation

- Birds migrate in response to species specific physiological conditions

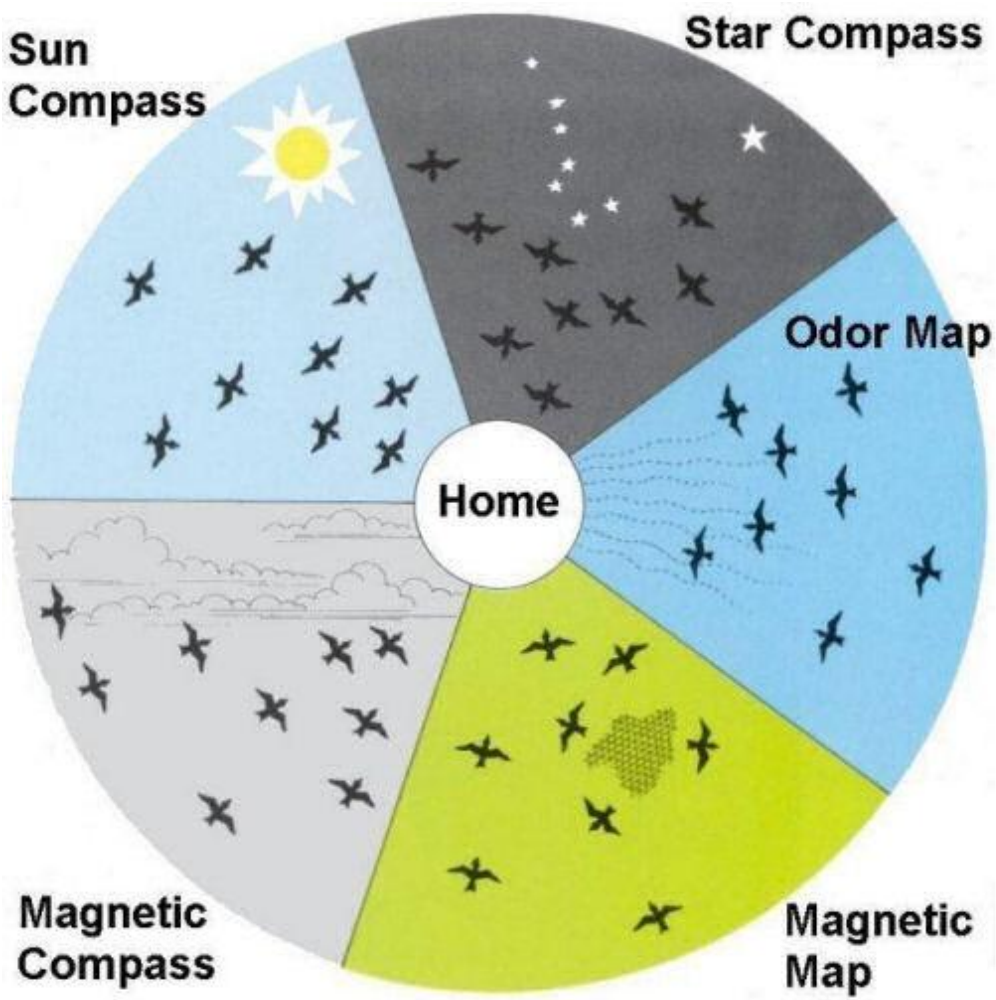
- The photoperiod is an important migratory cue for birds
  - Days shorter = shrinking of gonadal regression and storing of fat for migration south
  - Days longer = increase of gonadal development and storing of fat for migration north.

Migrate at day

Black geese flying in a V-formation

Migrate at night

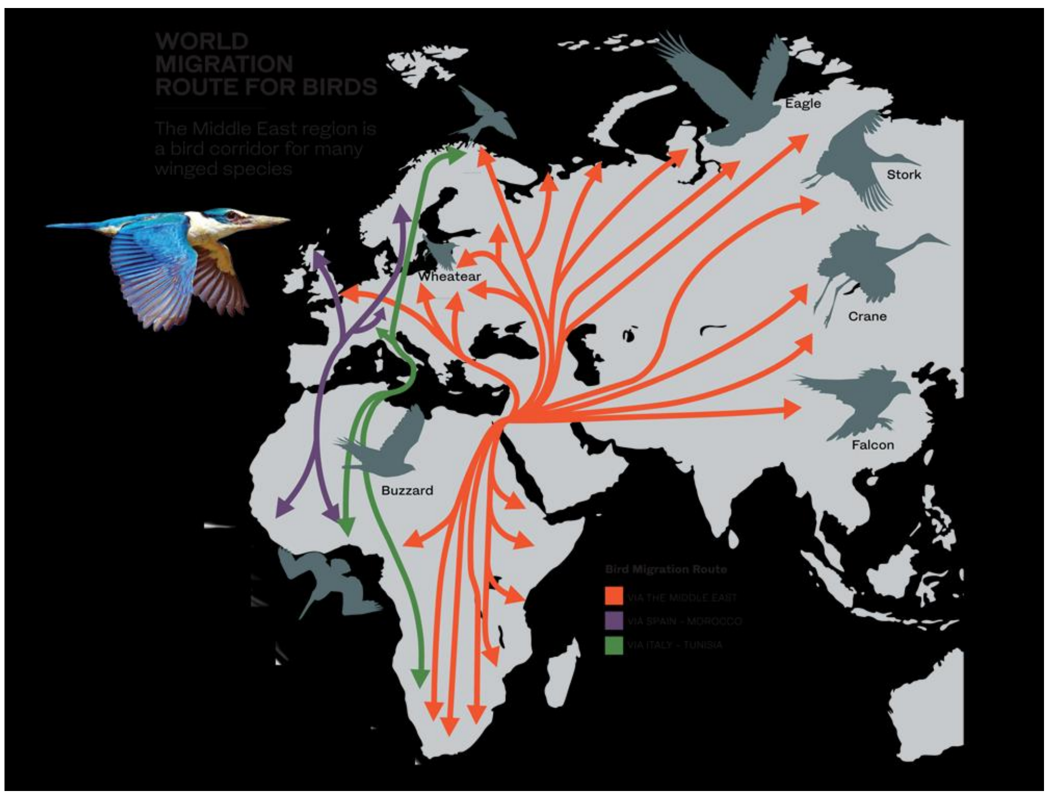
Major Concepts



- Migration and navigation allow birds to live, feed, and reproduce in environments favorable to the survival of adults and young.
- http://www.paulnoll.com/Oregon/Birds/migrate-navigation.html - website explains the mechanisms that drive migration.

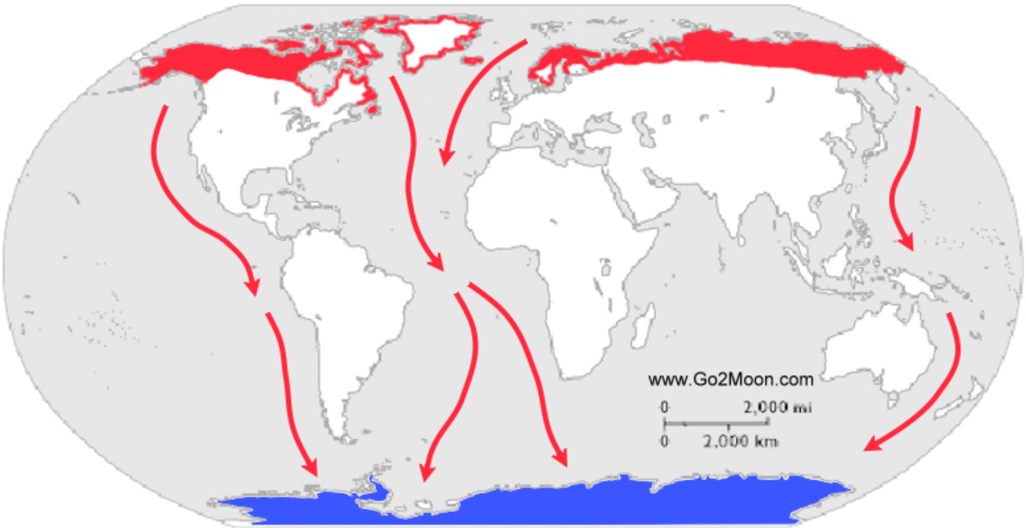
Navigation

- Birds use both route based and location based navigation.
- Route based navigation uses landmarks
- Location based navigation uses sun compass, celestial cues and the earth’s magnetic fields.



Arctic Turn

- Migratory route takes them from the Arctic to the Antarctic and back again each year, a distance of approximately 40,000km.
- Red = Breeding Grounds
- Blue = Wintering Grounds



White stork

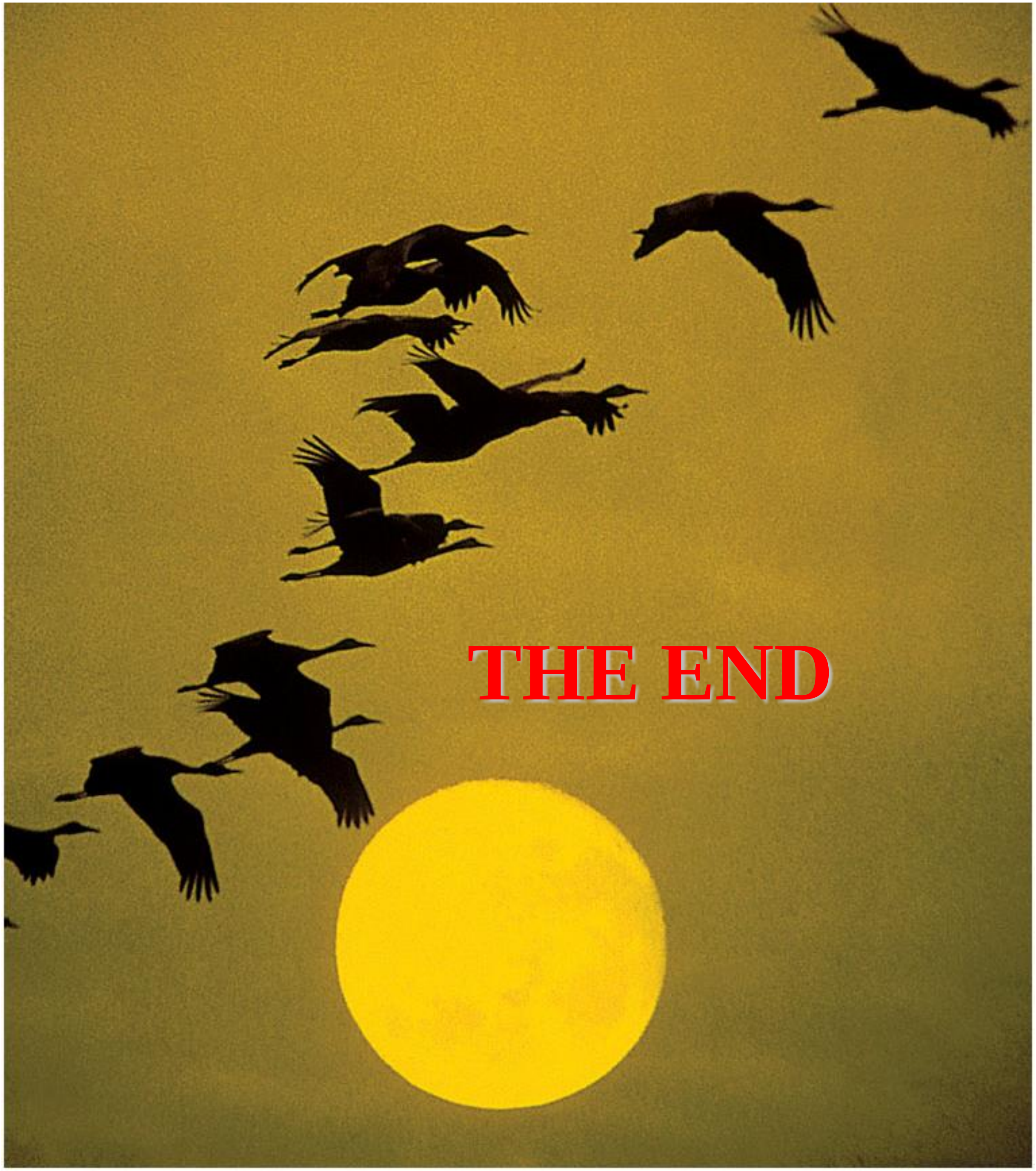
The white stork provides an example of how the amount of food available affects bird migration.

These storks used to come to the Iberian Peninsula to spend the summer and breed. Then, in the autumn, they migrated to Africa to winter there. However, more and more storks now winter on the Peninsula

This is due to:

A greater number of rubbish dumps that provide them with food throughout the year.

It is likely that the change is also related to the higher temperatures resulting from **climate change**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